

Geometry-based Methods in Vision

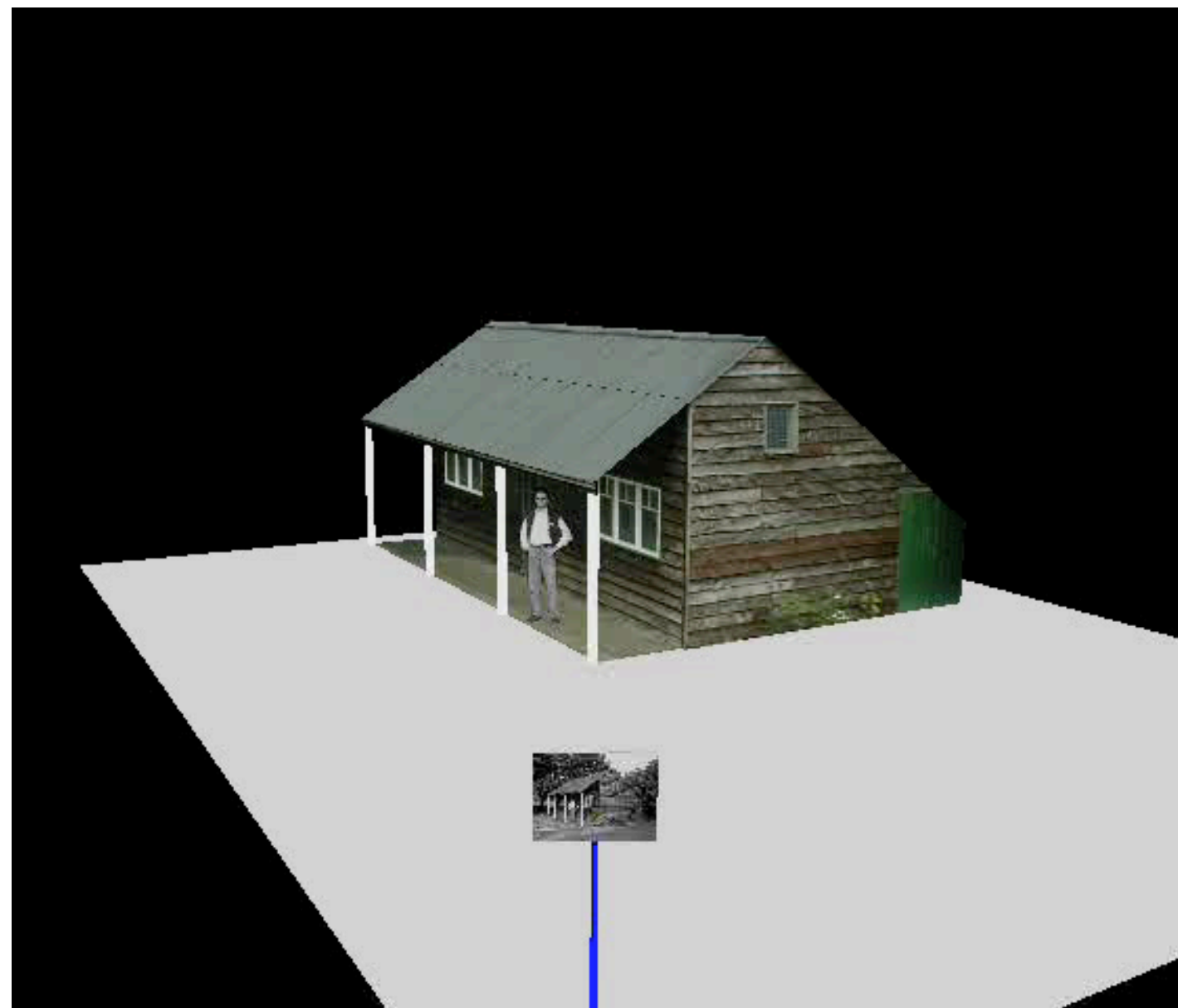
CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

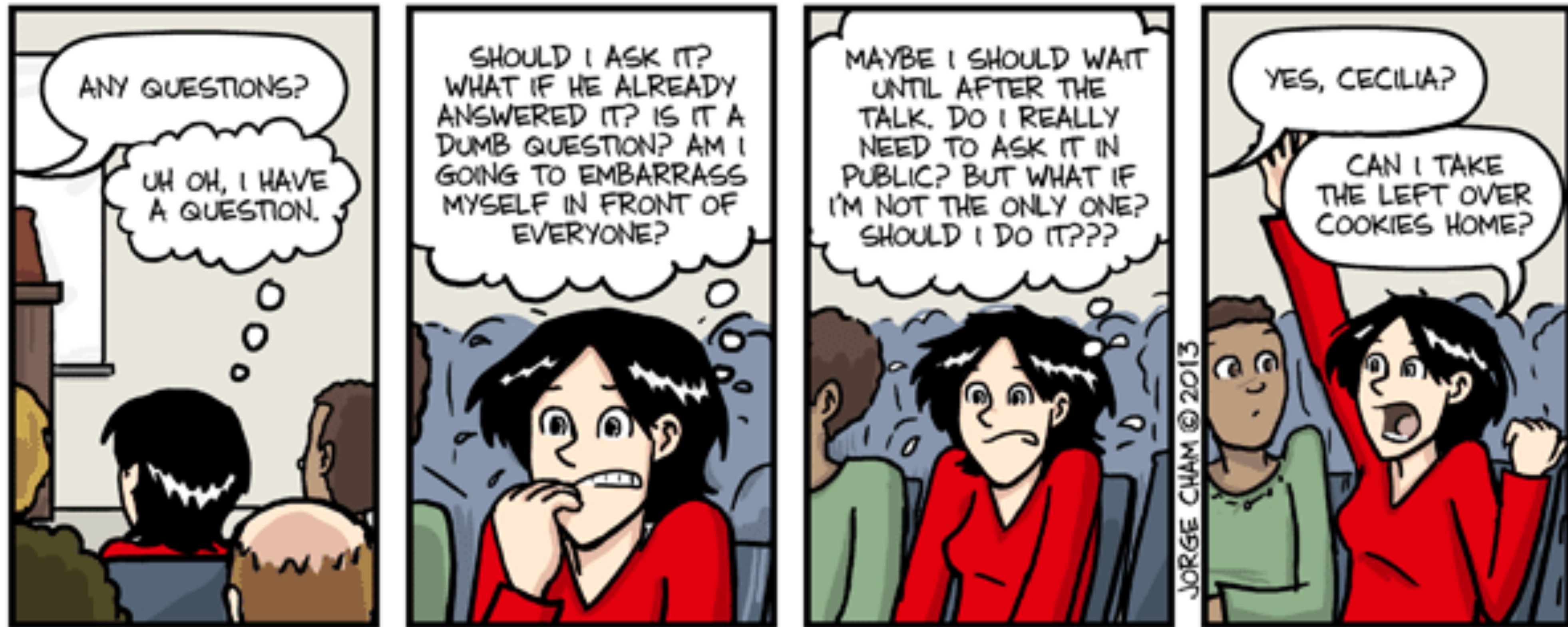
Administrivia

- Assignment 1
 - *Due on Tue next week*

Lecture 8: Single-view Reconstruction



A reminder to ask questions!

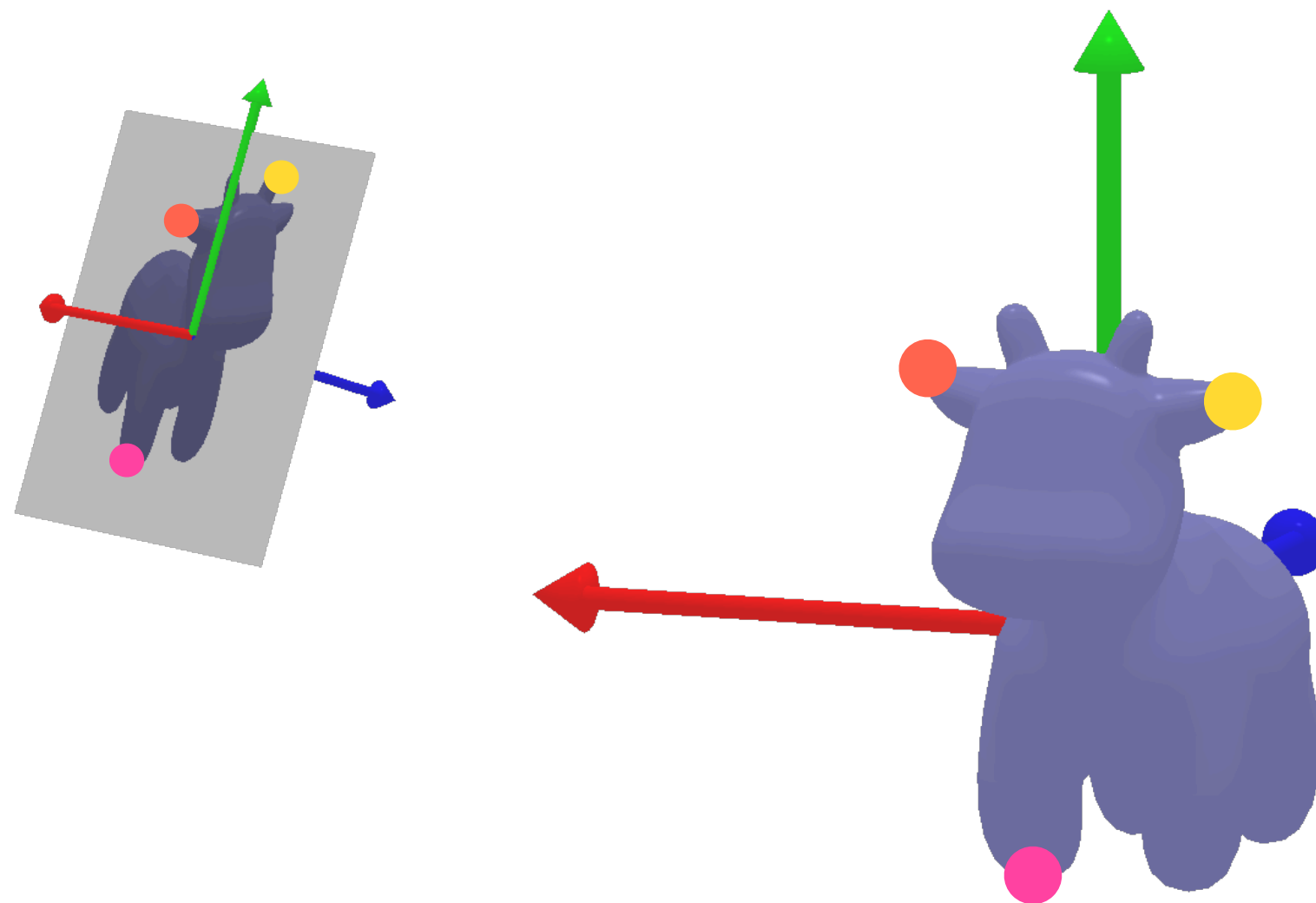


Determining the Camera Matrix

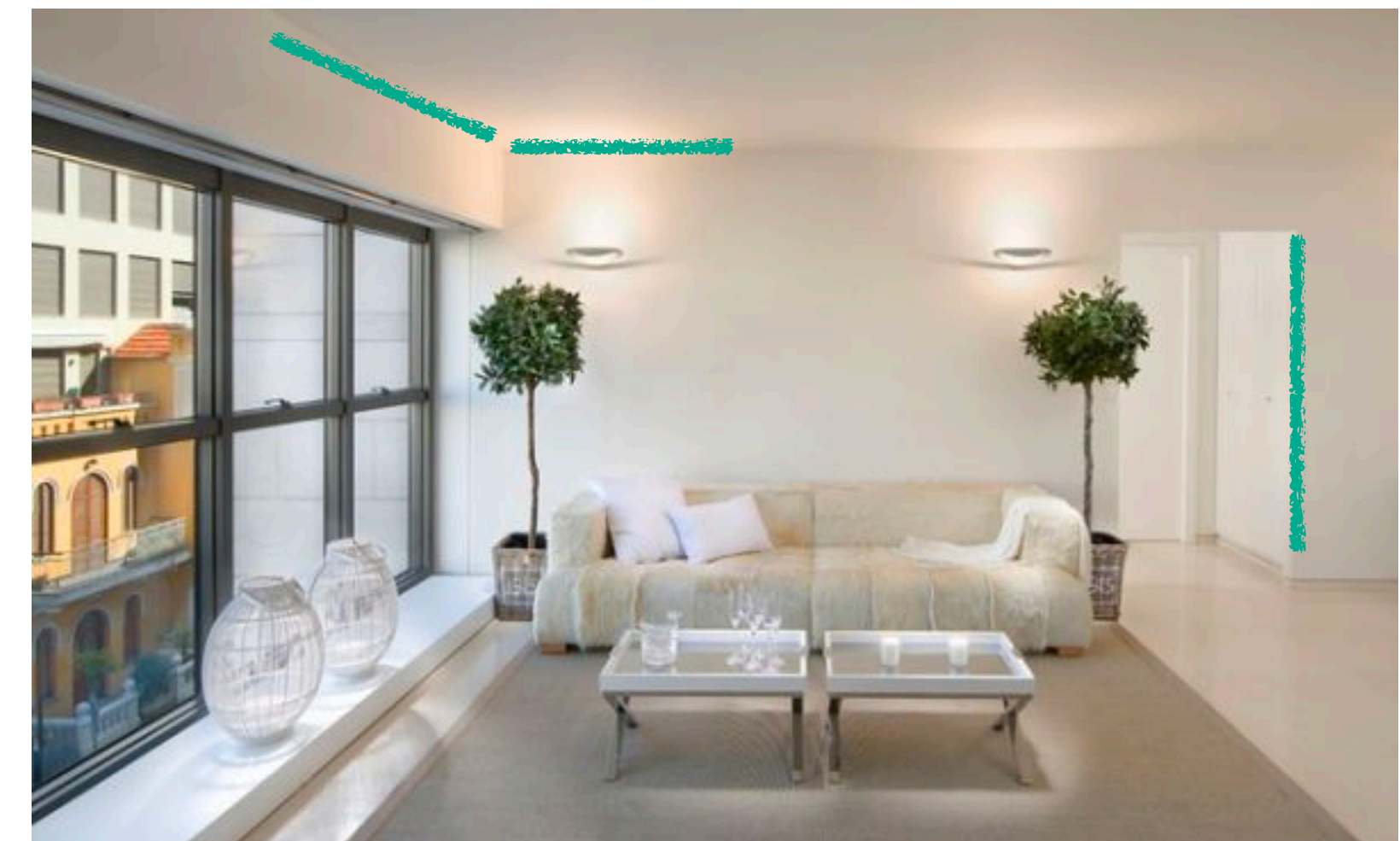
$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

Possible to recover intrinsics from annotation!



Can recover both, world to camera,
and camera to pixel transforms



What if the underlying 3D model is unknown?

need to introduce some concepts before

Direction Vectors and Projections

Q: Given an image coordinate $\mathbf{x}$, what is the direction of the corresponding ray?

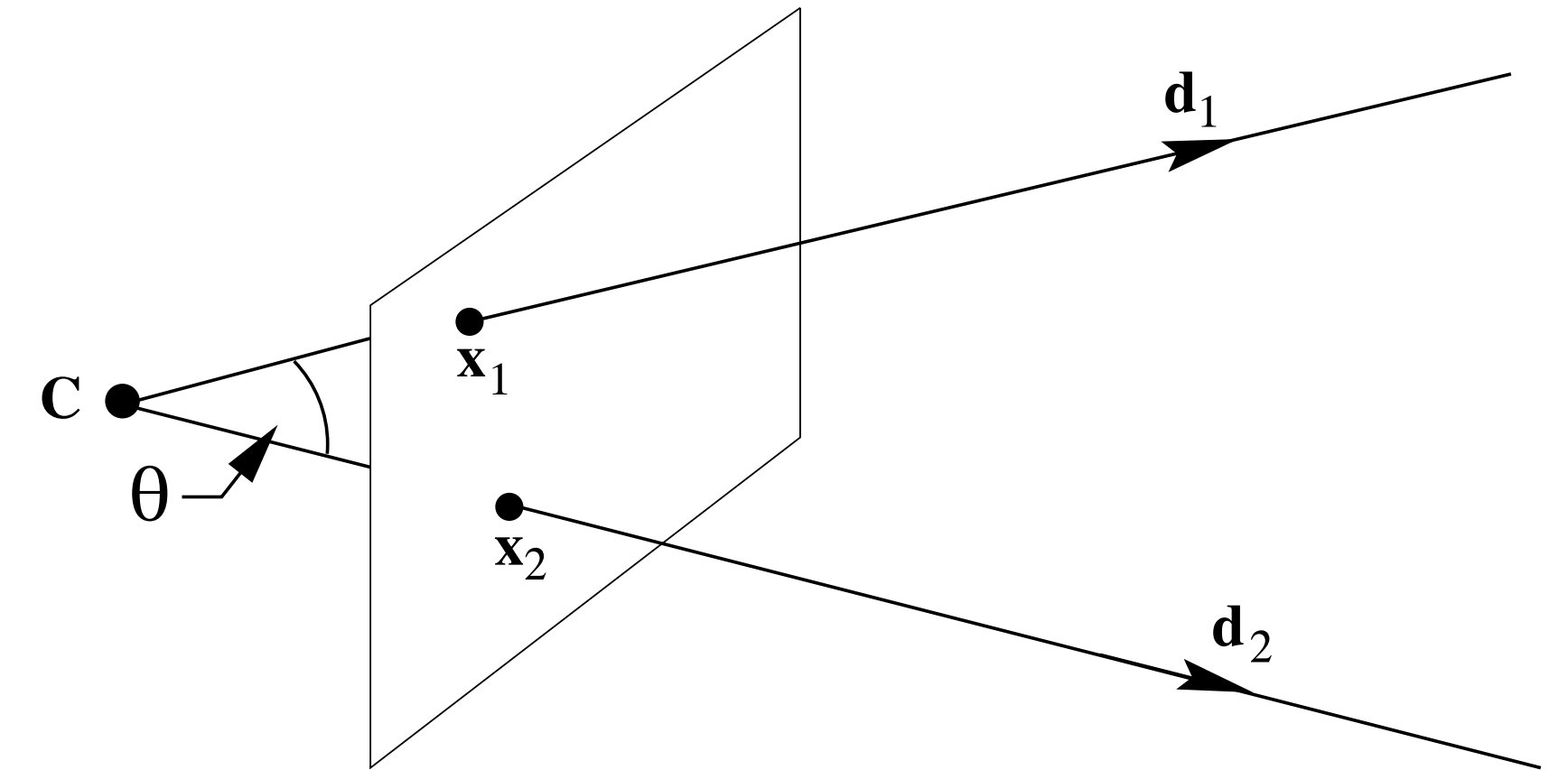
$$\mathbf{d} = R^T K^{-1} \mathbf{x} \quad (\text{not a normalized vector})$$

$$\mathbf{d} = K^{-1} \mathbf{x} \quad (\text{in camera frame})$$

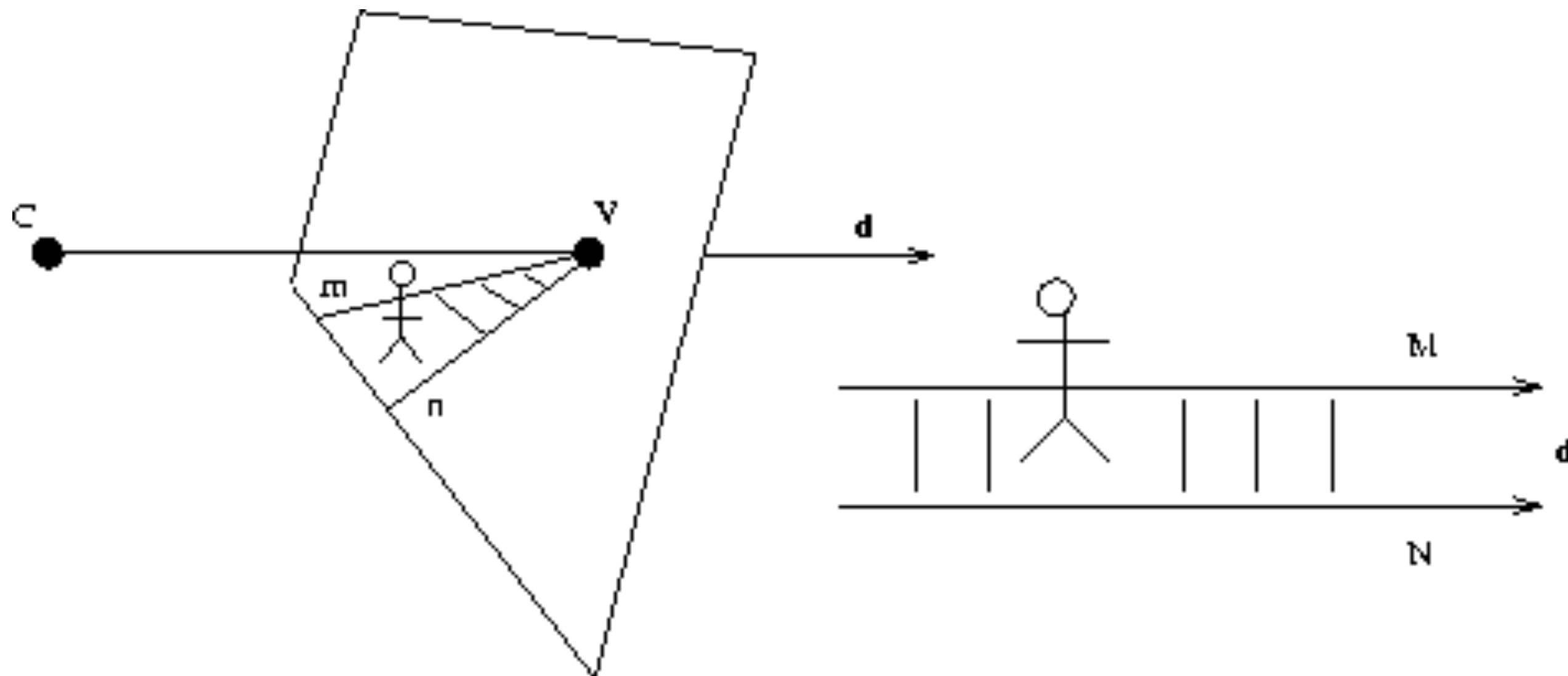
Q: Where does the point at infinity in direction $\mathbf{d}$ project?

$$\mathbf{x} = KR \begin{bmatrix} I & \tilde{\mathbf{C}} \end{bmatrix} \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix}$$

$$\mathbf{x} = KR \mathbf{d}$$



Direction from Vanishing Point



Q: Given vanishing point $\mathbf{v}$, what is the direction of the corresponding lines in 3D?

$$\mathbf{d} = R^T K^{-1} \mathbf{v}$$

$$M \equiv \begin{bmatrix} \tilde{\mathbf{x}} \\ 1 \end{bmatrix} + \begin{bmatrix} \lambda \mathbf{d} \\ 0 \end{bmatrix}$$

$$\lim_{\lambda \rightarrow \infty} \mathbf{P} \left(\begin{bmatrix} \tilde{\mathbf{x}} \\ 1 \end{bmatrix} + \begin{bmatrix} \lambda \mathbf{d} \\ 0 \end{bmatrix} \right) = \mathbf{v} \quad \mathbf{P} \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} = \mathbf{v}$$

Direction Vectors and Projections

Q: Given an image coordinate $\mathbf{x}$, what is the direction of the corresponding ray?

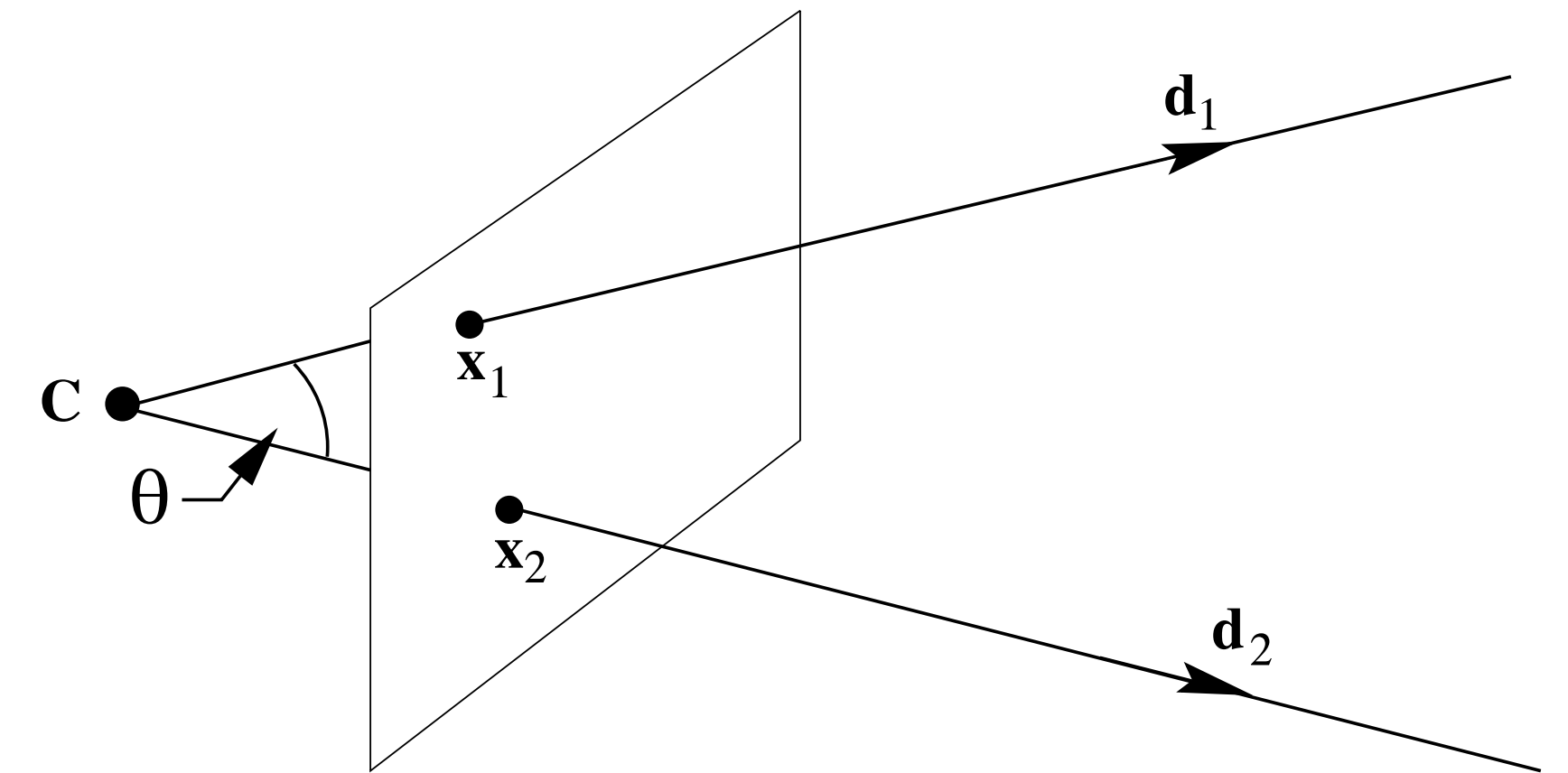
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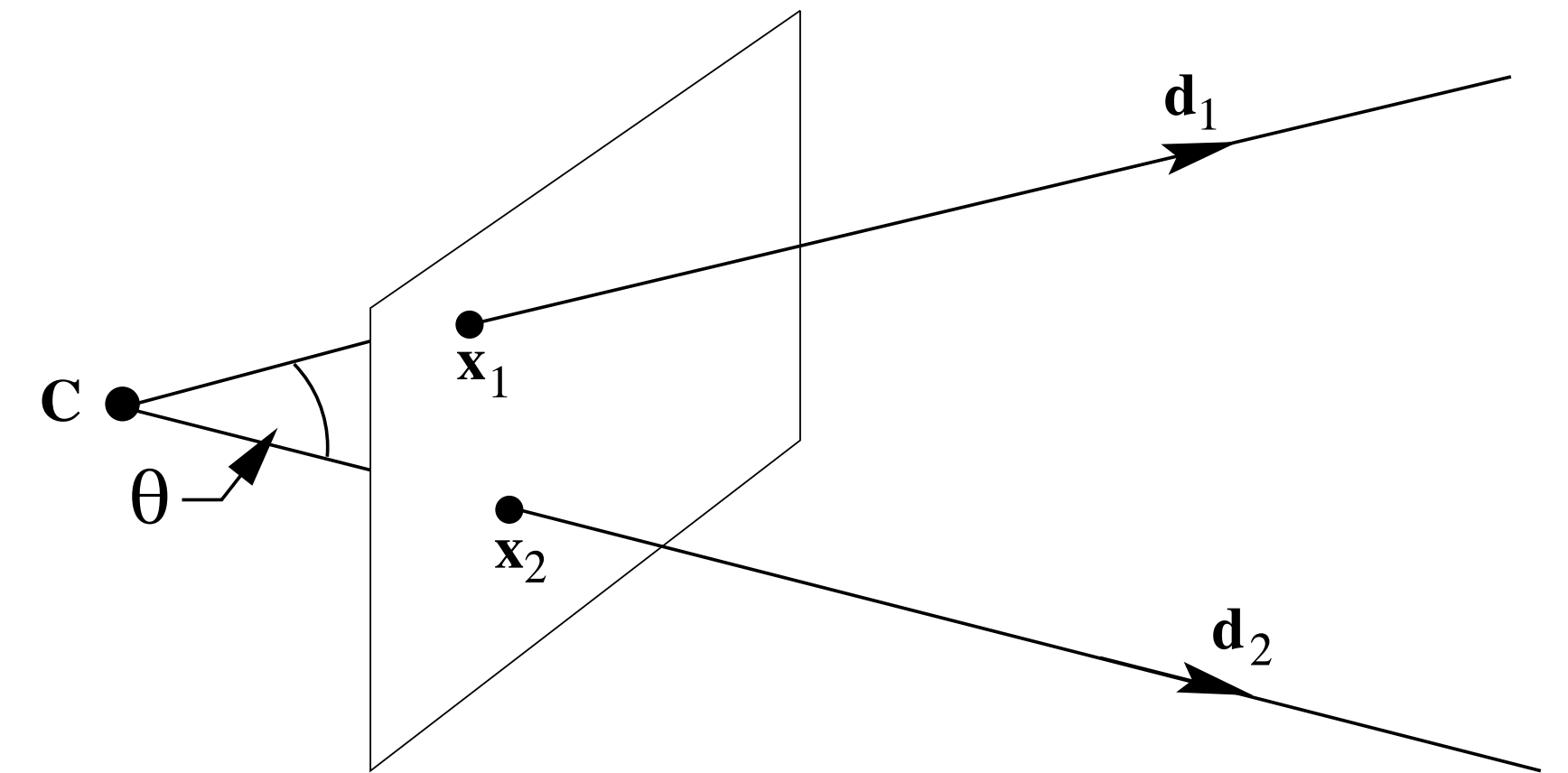


Direction Vectors and Projections

Q: Given points $\mathbf{x}_1$ and $\mathbf{x}_2$, what is angle between rays?

$$\begin{aligned} \cos(\theta) &= \frac{\mathbf{d}_1^\top \mathbf{d}_2}{\sqrt{\mathbf{d}_1^\top \mathbf{d}_1} \sqrt{\mathbf{d}_2^\top \mathbf{d}_2}} \\ &= \frac{\mathbf{x}_1^\top K^{-\top} R R^\top K^{-1} \mathbf{x}_2}{\sqrt{\dots} \sqrt{\dots}} = \frac{\mathbf{x}_1^\top K^{-\top} K^{-1} \mathbf{x}_2}{\sqrt{\dots} \sqrt{\dots}} \end{aligned}$$

Only depends on the camera intrinsics!



$$\mathbf{d} = R^\top K^{-1} \mathbf{x} \quad \mathbf{x} = K R \mathbf{d}$$

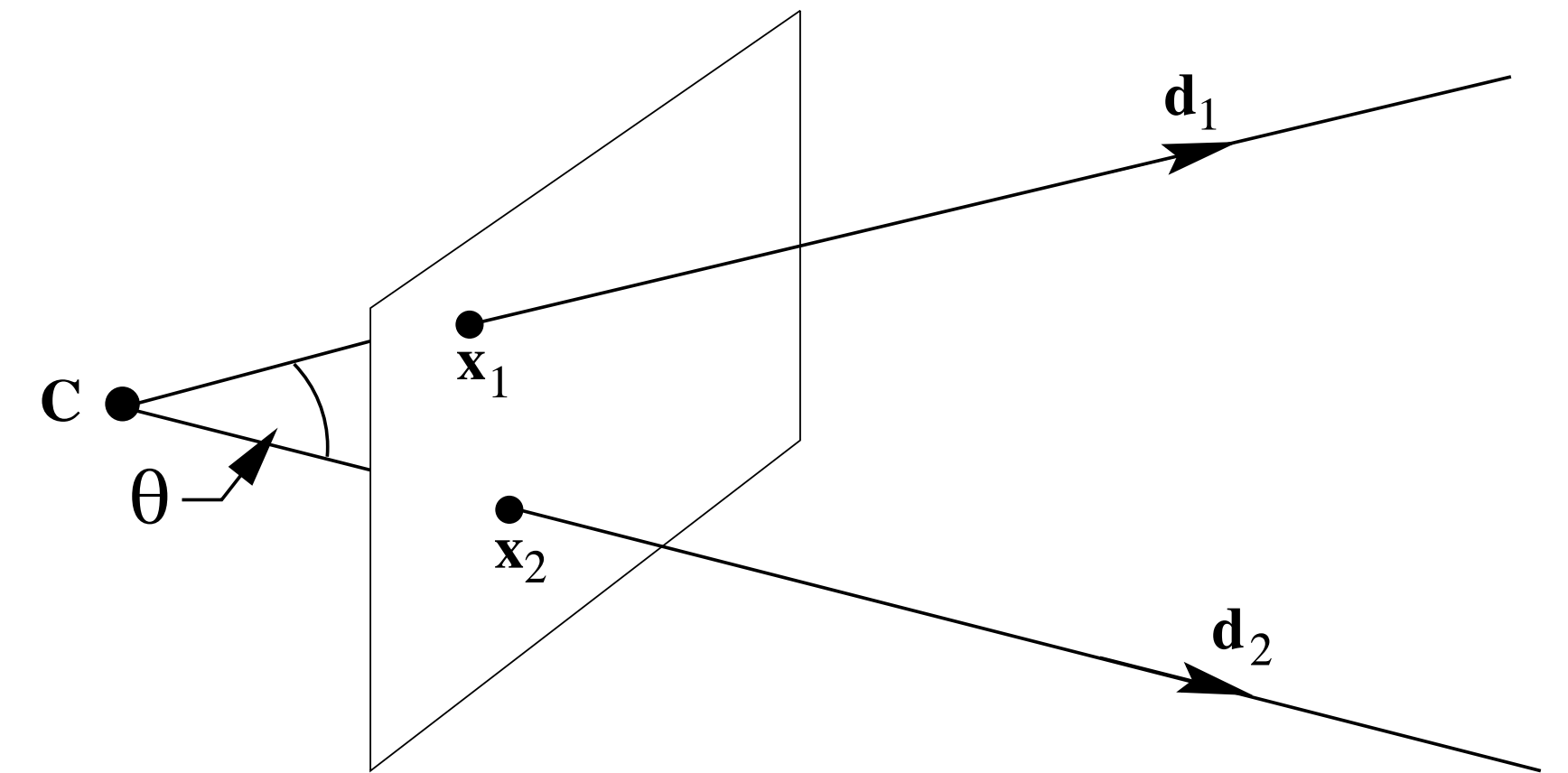
Direction Vectors and Projections

Q: Given points $\mathbf{x}_1$ and $\mathbf{x}_2$, what is angle between rays?

$$\cos(\theta) = \frac{\mathbf{d}_1^T \mathbf{d}_2}{\sqrt{\mathbf{d}_1^T \mathbf{d}_1} \sqrt{\mathbf{d}_2^T \mathbf{d}_2}}$$

$$= \frac{\mathbf{x}_1^T \omega \mathbf{x}_2}{\sqrt{\dots} \sqrt{\dots}}$$

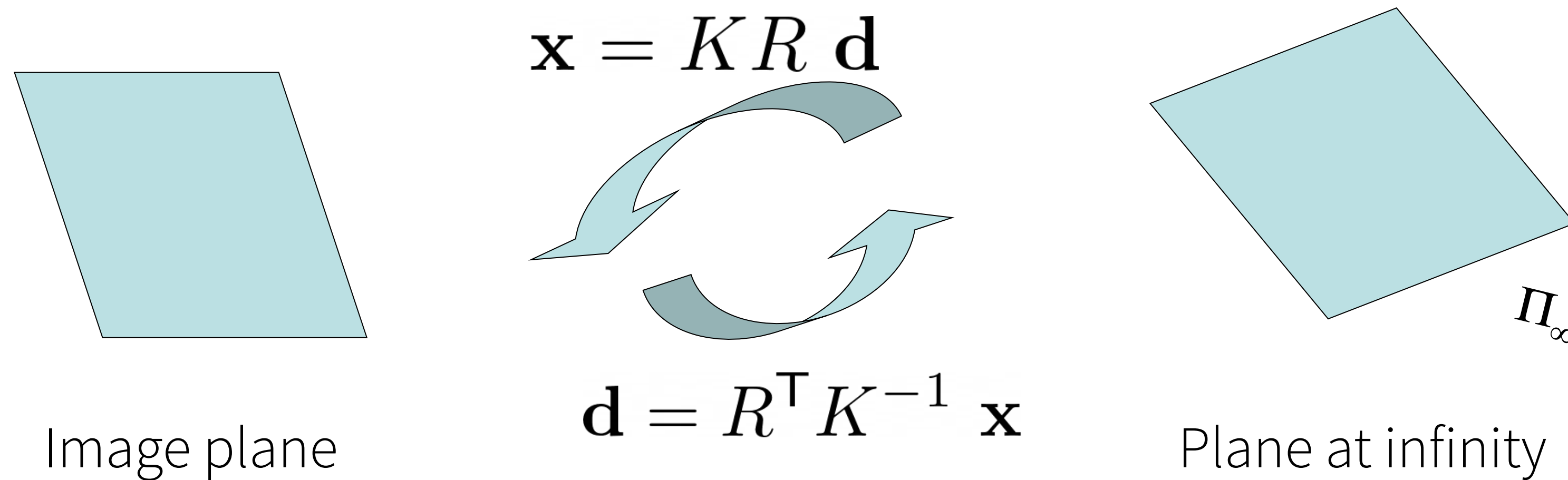
$$K^{-T} K^{-1} = \omega$$



$$\mathbf{d} = R^T K^{-1} \mathbf{x} \quad \mathbf{x} = K R \mathbf{d}$$

Related to a special conic on plane at infinity

Image plane and the Plane at Infinity



There is a homography **H** that relates the two
(independent of camera location)

Image plane and the Plane at Infinity

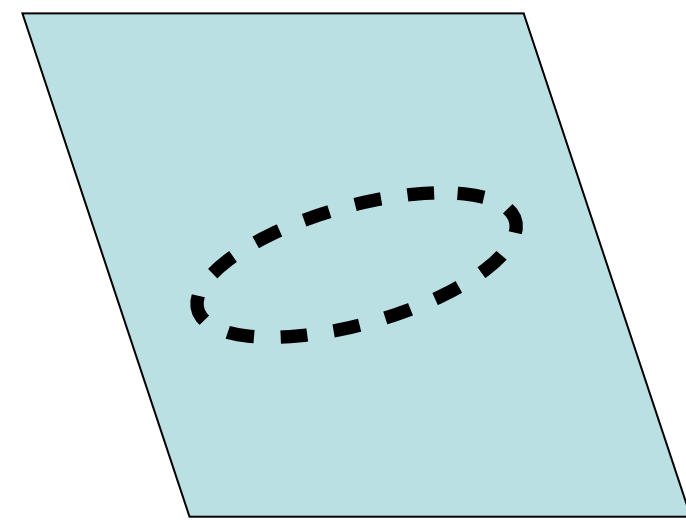
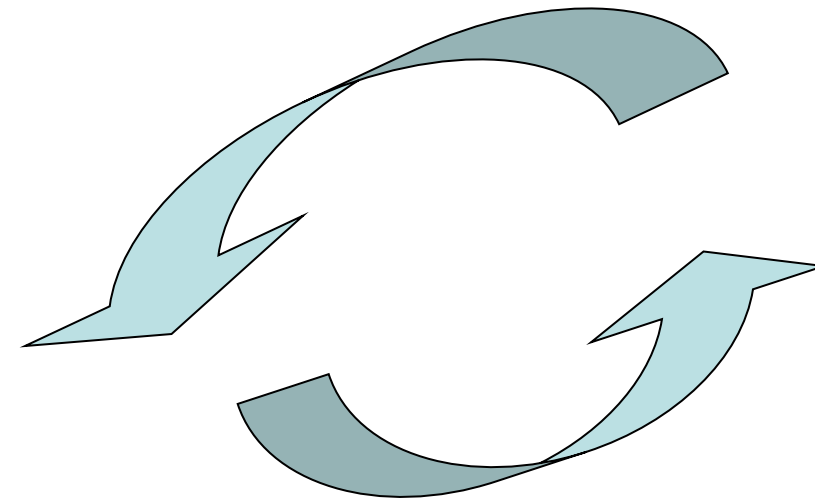
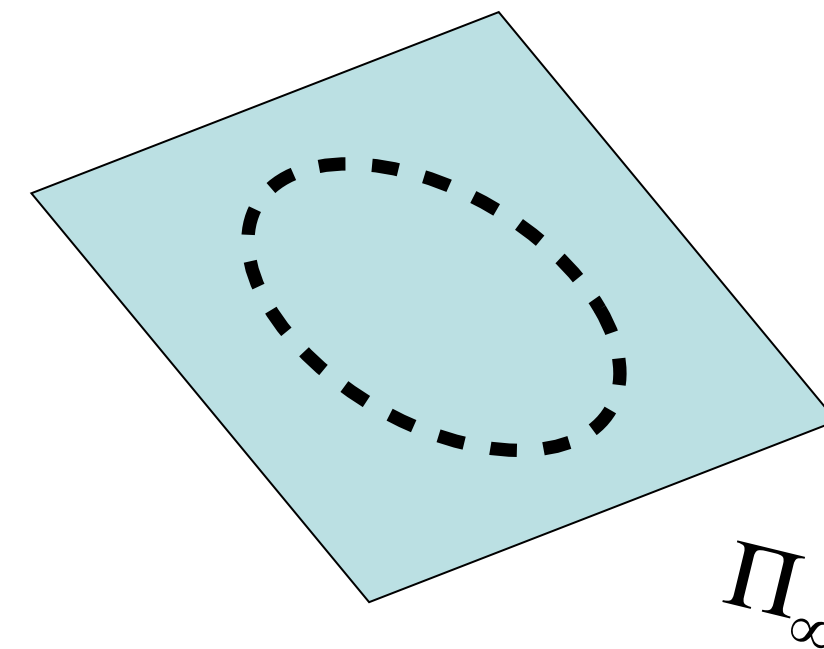


Image plane

$$\mathbf{x} = K R \mathbf{d}$$



$$\mathbf{d} = R^T K^{-1} \mathbf{x}$$



Plane at infinity

Allows computing the
'image' of any conic on
the plane at infinity

$$\mathbf{C}' = \mathbf{H}^{-T} \mathbf{C} \mathbf{H}^{-1}$$

Absolute Conic — Analogue of Circular points in 3D

$\mathbb{P}^3$

All spheres intersect the plane at infinity at the ‘absolute conic’

$$\mathbf{X} \equiv [x, y, z, w]^T \text{ s.t. } x^2 + y^2 + z^2 = 0; w = 0$$

A conic on the plane at infinity!

$$\mathbf{X} \equiv [\mathbf{d}, 0]^T \text{ s.t. } \mathbf{d}^T I \mathbf{d} = 0$$

$\mathbb{P}^2$

All circles intersect the line at infinity at ‘circular points’

$$\mathbf{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \quad \mathbf{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

Alternate definition:

$$\mathbf{x} \equiv [x, y, w]^T \text{ s.t. } x^2 + y^2 = 0; w = 0$$

(just so happens there are only 2 unique solutions)

Absolute Conic Ω

$$\mathbf{X} \equiv [x, y, z, w]^T \text{ s.t.} \\ x^2 + y^2 + z^2 = 0; w = 0$$

$$\mathbf{X} \equiv [\mathbf{d}, 0]^T \text{ s.t.} \\ \mathbf{d}^T I \mathbf{d} = 0$$

All spheres intersect the plane at infinity at
the ‘absolute conic’

$$ax^2 + ay^2 + az^2 + bxw + dyw + fzw + w^2 = 0$$

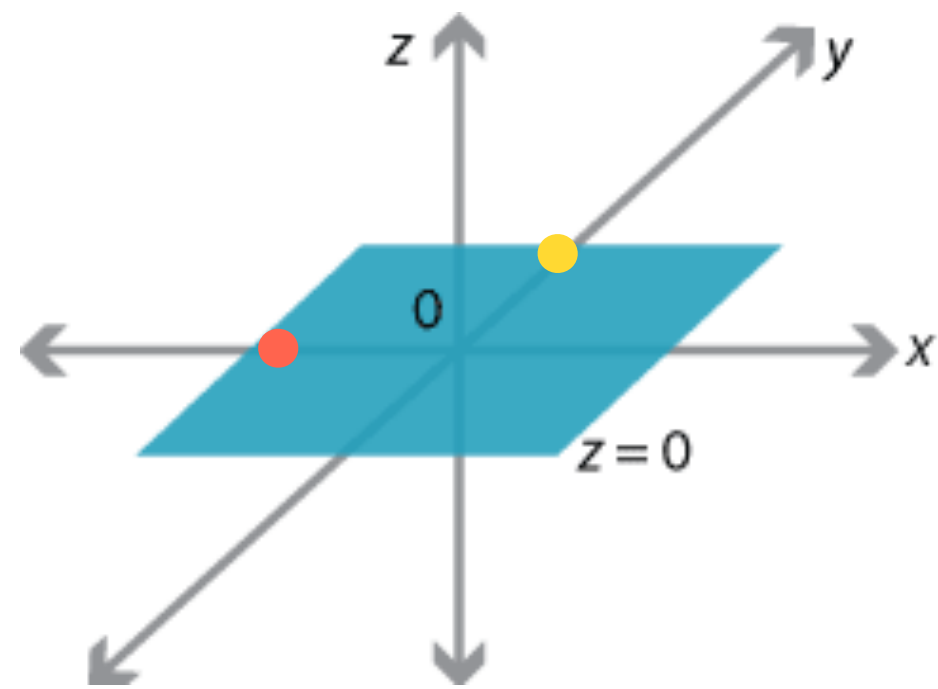
$$w = 0 \rightarrow x^2 + y^2 + z^2 = 0$$

Absolute Conic Ω

$$\mathbf{X} \equiv [x, y, z, w]^T \text{ s.t. } x^2 + y^2 + z^2 = 0; w = 0$$

$$\mathbf{X} \equiv [\mathbf{d}, 0]^T \text{ s.t. } \mathbf{d}^T I \mathbf{d} = 0$$

All circles on any plane π intersect Ω
in the 2 circular points of the plane



Consider plane $z=0$

All circles pass:

$$\mathbf{I} = \begin{bmatrix} 1 \\ i \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{J} = \begin{bmatrix} 1 \\ -i \\ 0 \\ 0 \end{bmatrix}$$

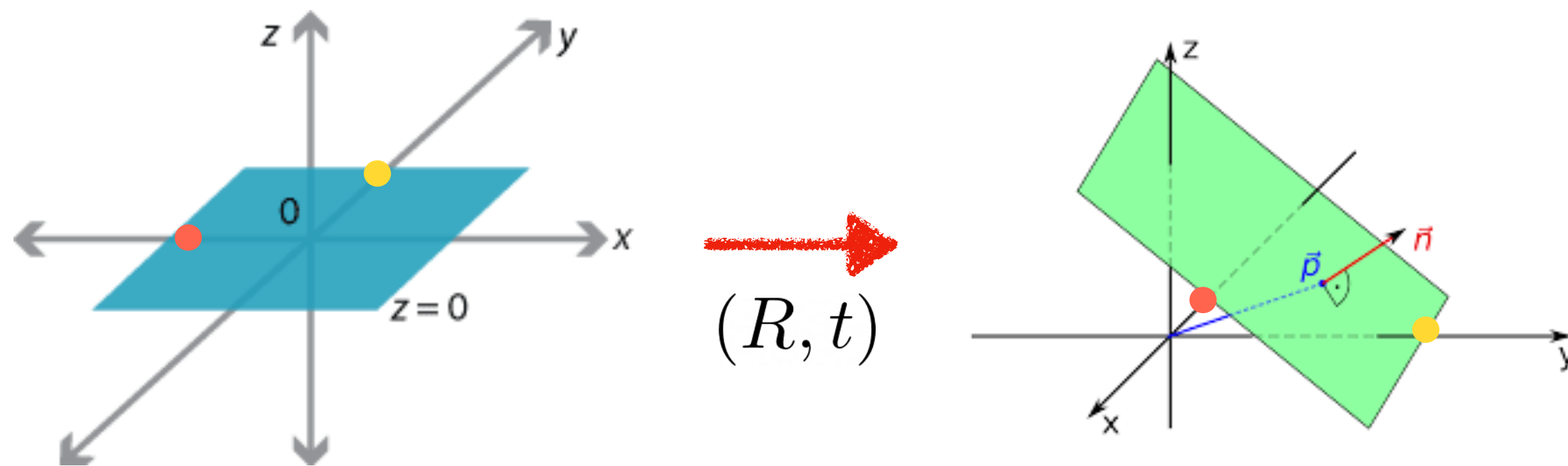
$\mathbf{I}, \mathbf{J}$ lie on Ω

Absolute Conic Ω

$$\mathbf{X} \equiv [x, y, z, w]^T \text{ s.t. } x^2 + y^2 + z^2 = 0; w = 0$$

$$\mathbf{X} \equiv [\mathbf{d}, 0]^T \text{ s.t. } \mathbf{d}^T I \mathbf{d} = 0$$

All circles on any plane π intersect Ω
in the 2 circular points of the plane



Arbitrary plane π

$$\mathbf{I}_\pi \equiv [R \ t] \mathbf{I}$$

$$\mathbf{J}_\pi \equiv [R \ t] \mathbf{J}$$

Also lie on Ω

$$\mathbf{I} = \begin{bmatrix} 1 \\ i \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{J} = \begin{bmatrix} 1 \\ -i \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{I}, \mathbf{J}$ lie on Ω

Image of the Absolute Conic (IAC): ω

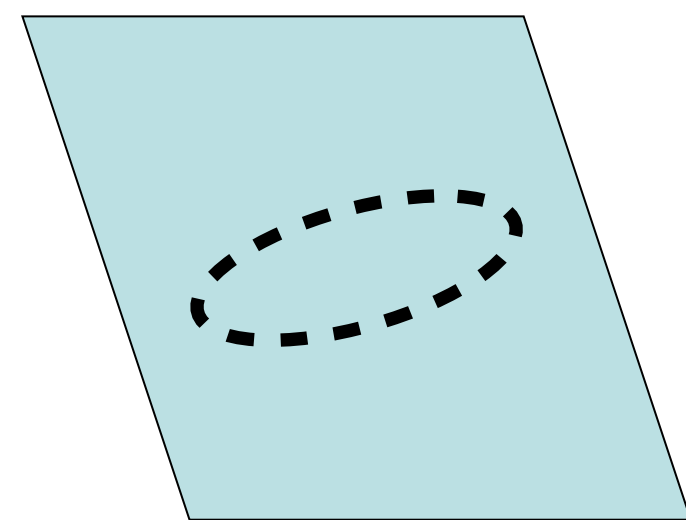
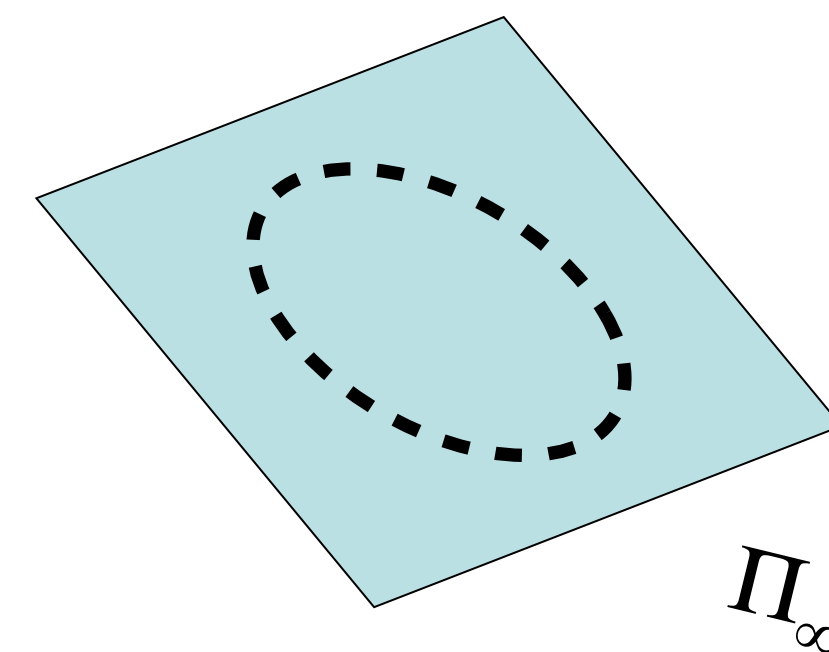


Image plane

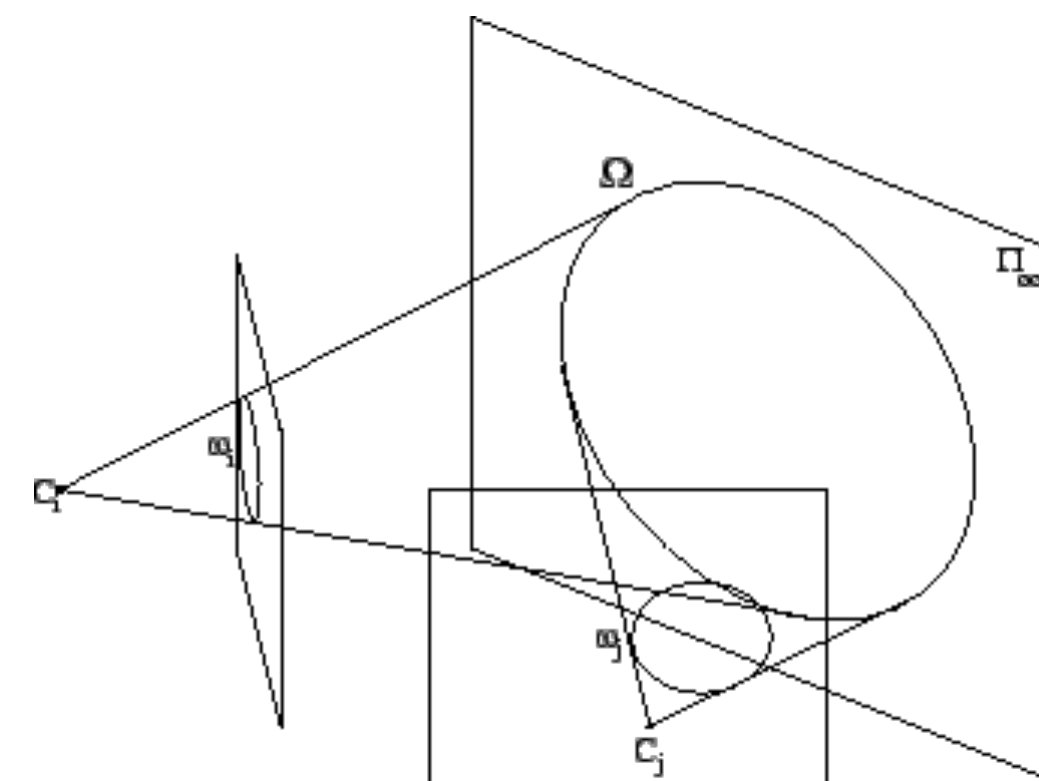
$$\mathbf{x} = KR \mathbf{d}$$

$$C' = H^{-T}CH^{-1}$$



Plane at infinity

$$\begin{aligned}\omega &= (KR)^{-T}I(KR)^{-1} \\ &= K^{-T}RR^{-T}K^{-1} \\ &= K^{-T}K^{-1}\end{aligned}$$

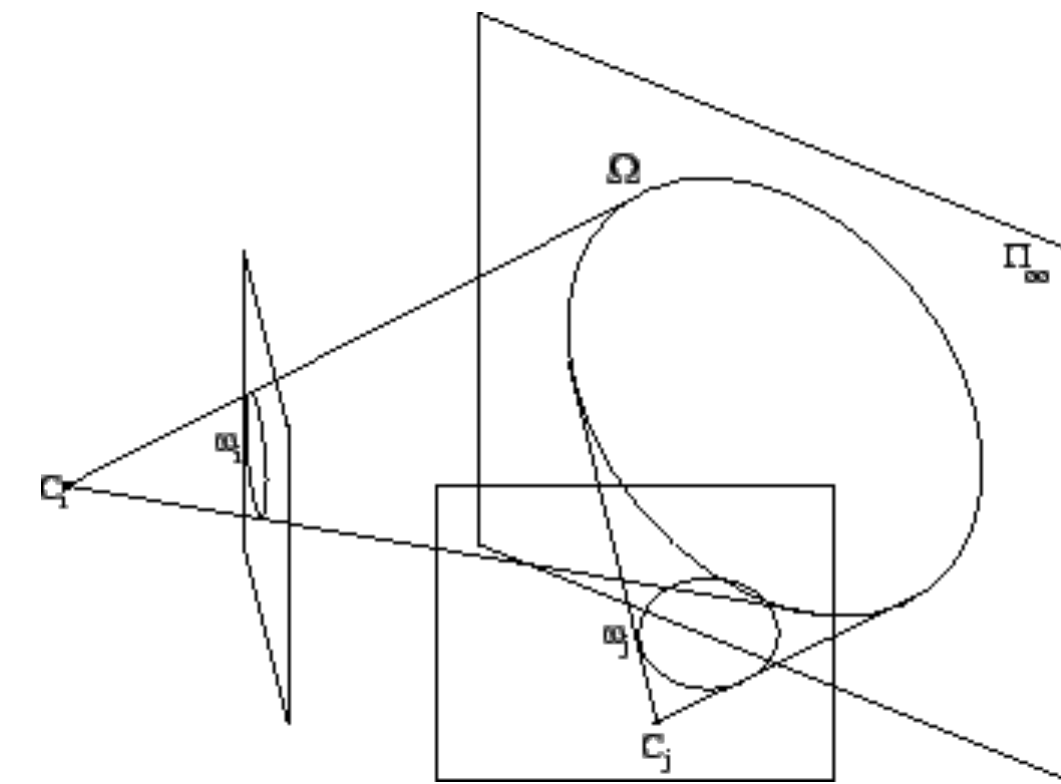


$$\mathbf{d}^T I \mathbf{d} = 0$$

Image of the Absolute Conic (IAC): ω

$$\omega = K^{-T} K^{-1}$$

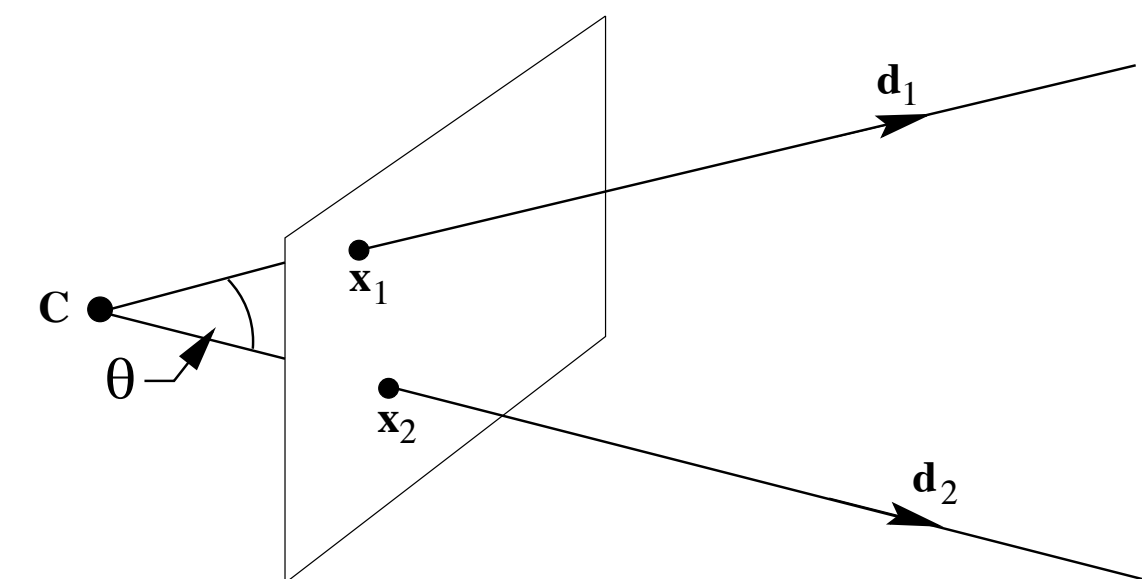
Depends **only** on the camera intrinsic
 can compute K if we know IAC
 (Cholesky decomposition)



$$\mathbf{d}^T I \mathbf{d} = 0$$

How can we determine IAC?

$$\cos(\theta) = \frac{\mathbf{d}_1^T \mathbf{d}_2}{\sqrt{\mathbf{d}_1^T \mathbf{d}_1} \sqrt{\mathbf{d}_2^T \mathbf{d}_2}} = \frac{\mathbf{x}_1^T \omega \mathbf{x}_2}{\sqrt{\dots} \sqrt{\dots}}$$



Option 1:

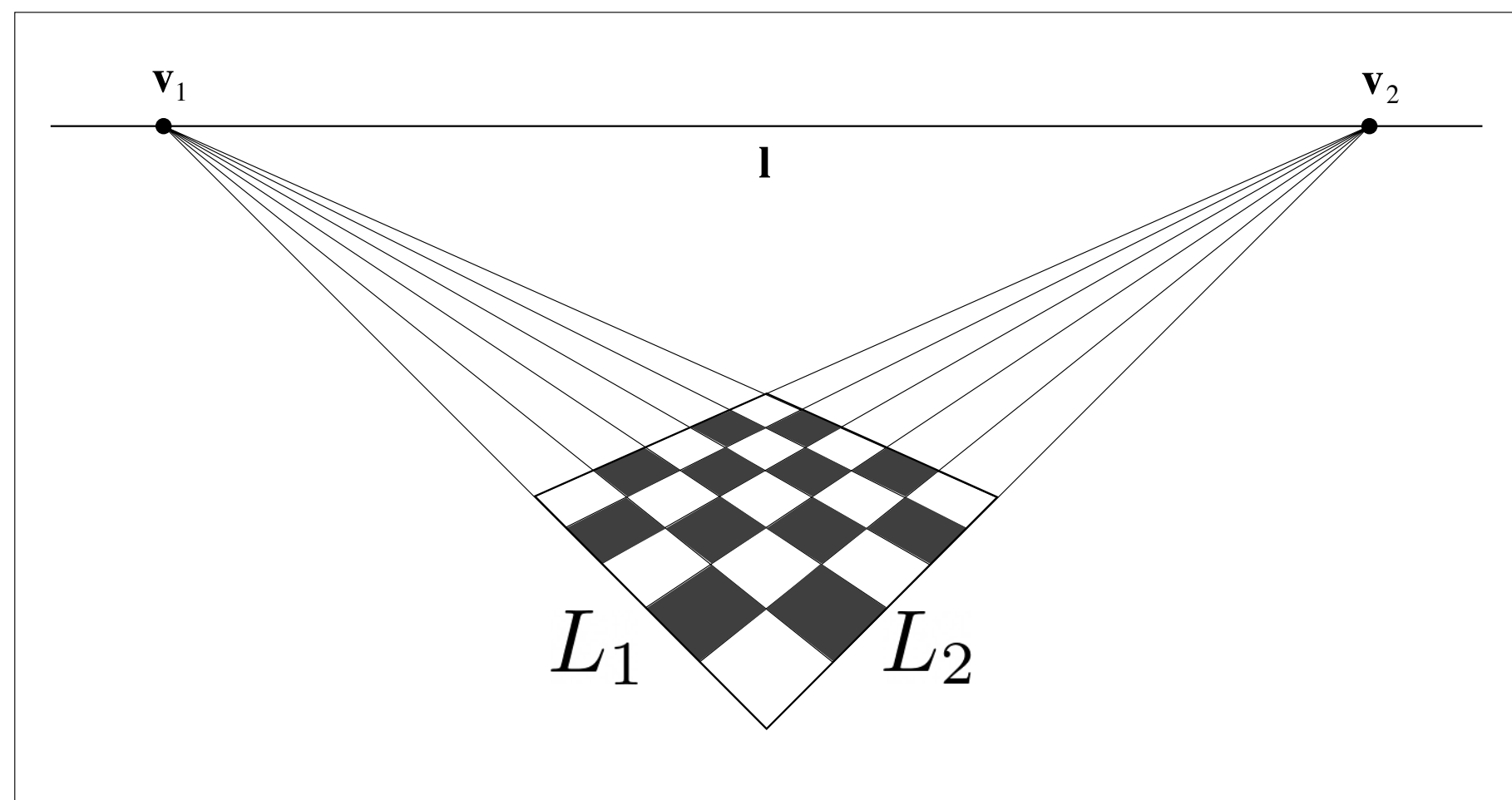
$$\mathbf{d}_1 \perp \mathbf{d}_2 \rightarrow \mathbf{x}_1^T \omega \mathbf{x}_2 = 0$$

Image of the Absolute Conic (IAC): ω

How can we determine IAC?

Option 1:

$$\mathbf{d}_1 \perp \mathbf{d}_2 \rightarrow \mathbf{x}_1^T \omega \mathbf{x}_2 = 0$$



If we can identify L_1 and L_2 whose directions are orthogonal:

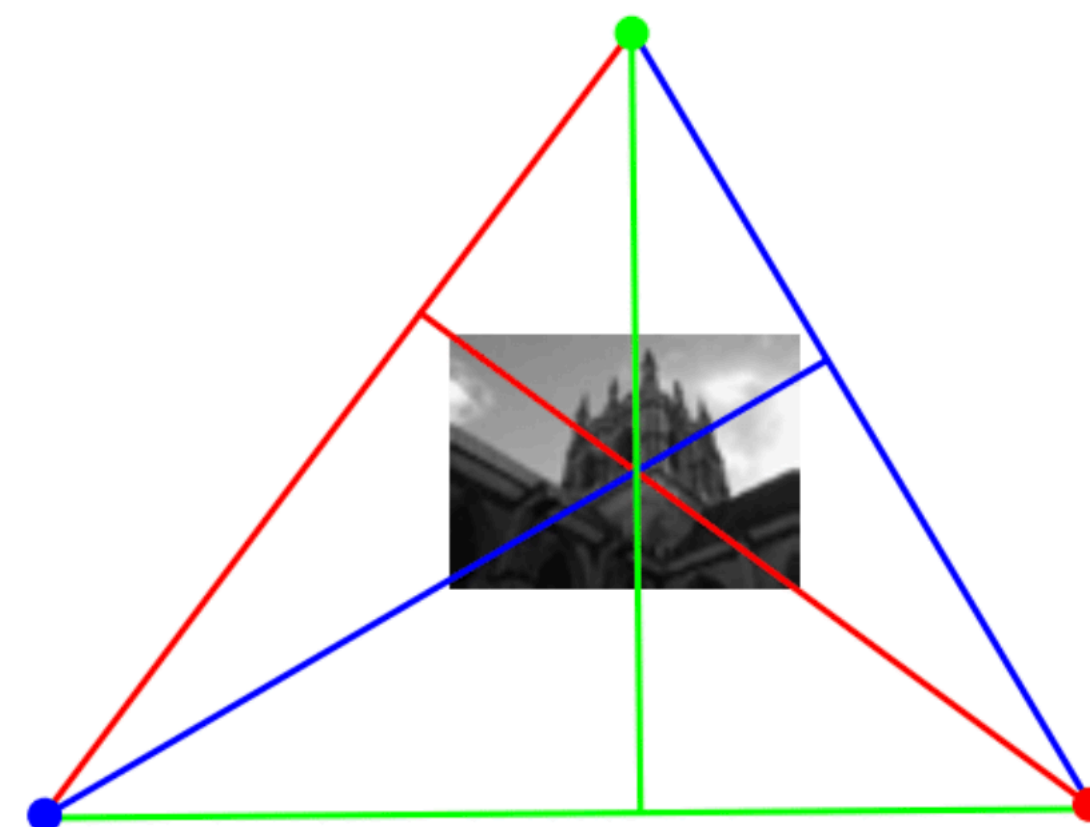
$$\mathbf{v}_1^T \omega \mathbf{v}_2 = 0$$

Note: unlike 2D rectification, simply annotating a pair of perpendicular lines is **not** enough!

We need to also identify their vanishing points!

Image of the Absolute Conic (IAC): ω

How can we determine IAC?



$$\mathbf{v}_i^T \omega \mathbf{v}_j = 0$$

Step 1: Annotate some parallel lines to get vanishing points (as these let us compute line directions)

Step 2: Identify sets of lines with perpendicular directions

Step 3: Each pair of vanishing points for perpendicular directions gives one constraint

Q: How many constraints are needed?

Image of the Absolute Conic (IAC): ω

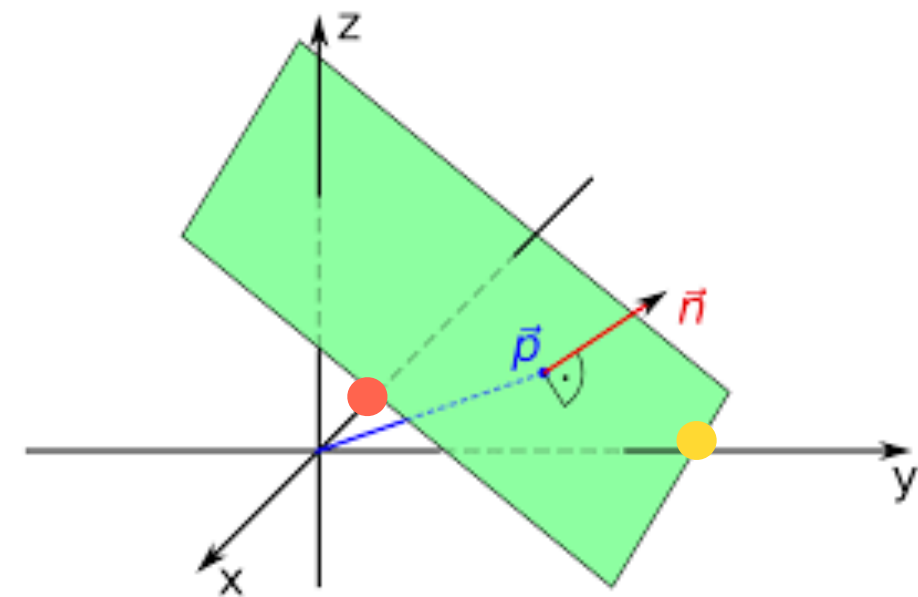
How can we determine IAC?

5 constraints needed

Option 1:

$$\mathbf{d}_1 \perp \mathbf{d}_2 \rightarrow \mathbf{x}_1^T \omega \mathbf{x}_2 = 0$$

Option 2:



$$\mathbf{I}_\pi \equiv \begin{bmatrix} R & t \end{bmatrix} \mathbf{I}$$

$$\mathbf{J}_\pi \equiv \begin{bmatrix} R & t \end{bmatrix} \mathbf{J}$$

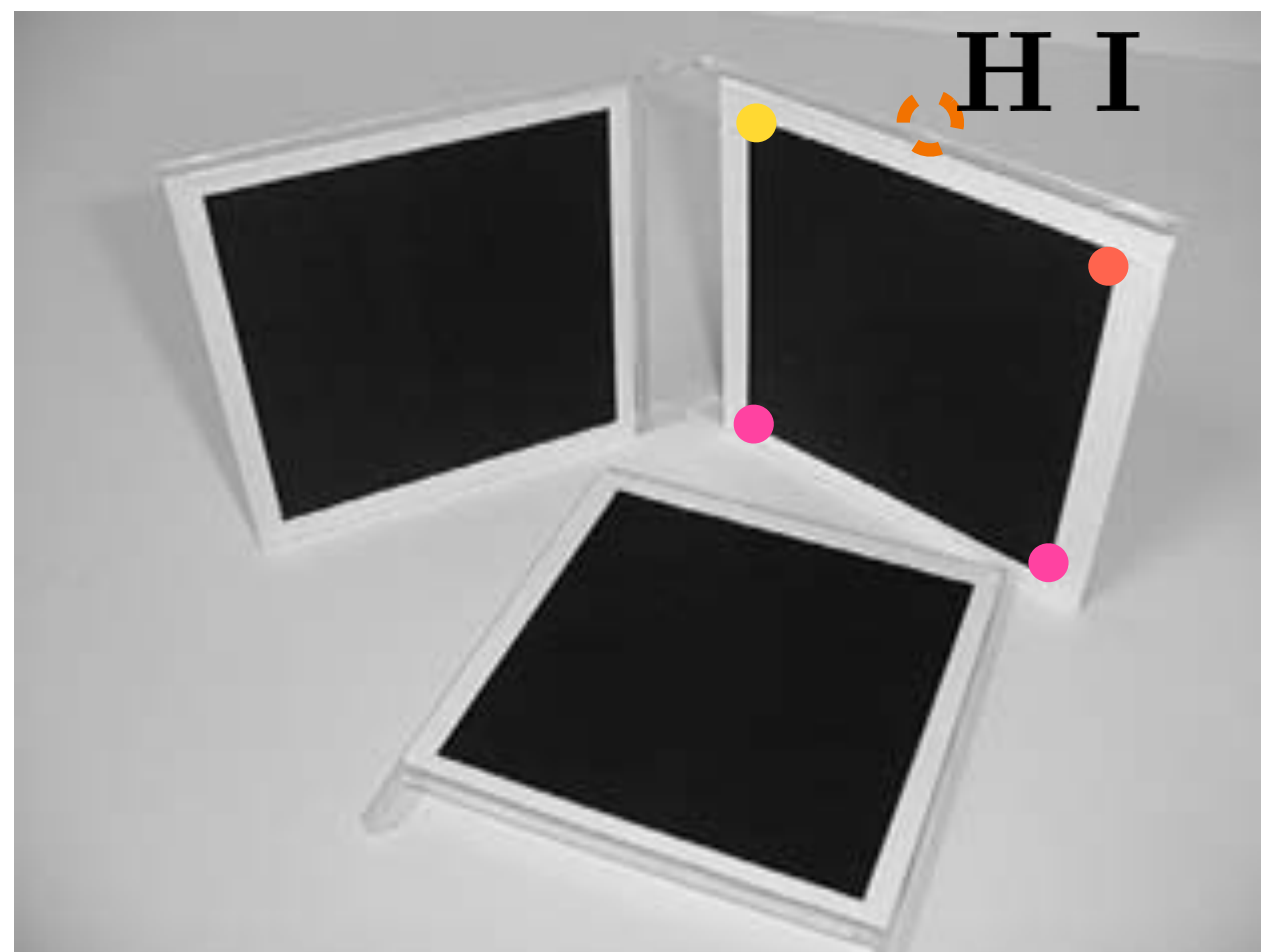
Arbitrary plane π

Also lie on Ω

Identify images of circular points of some plane
The image should lie on IAC

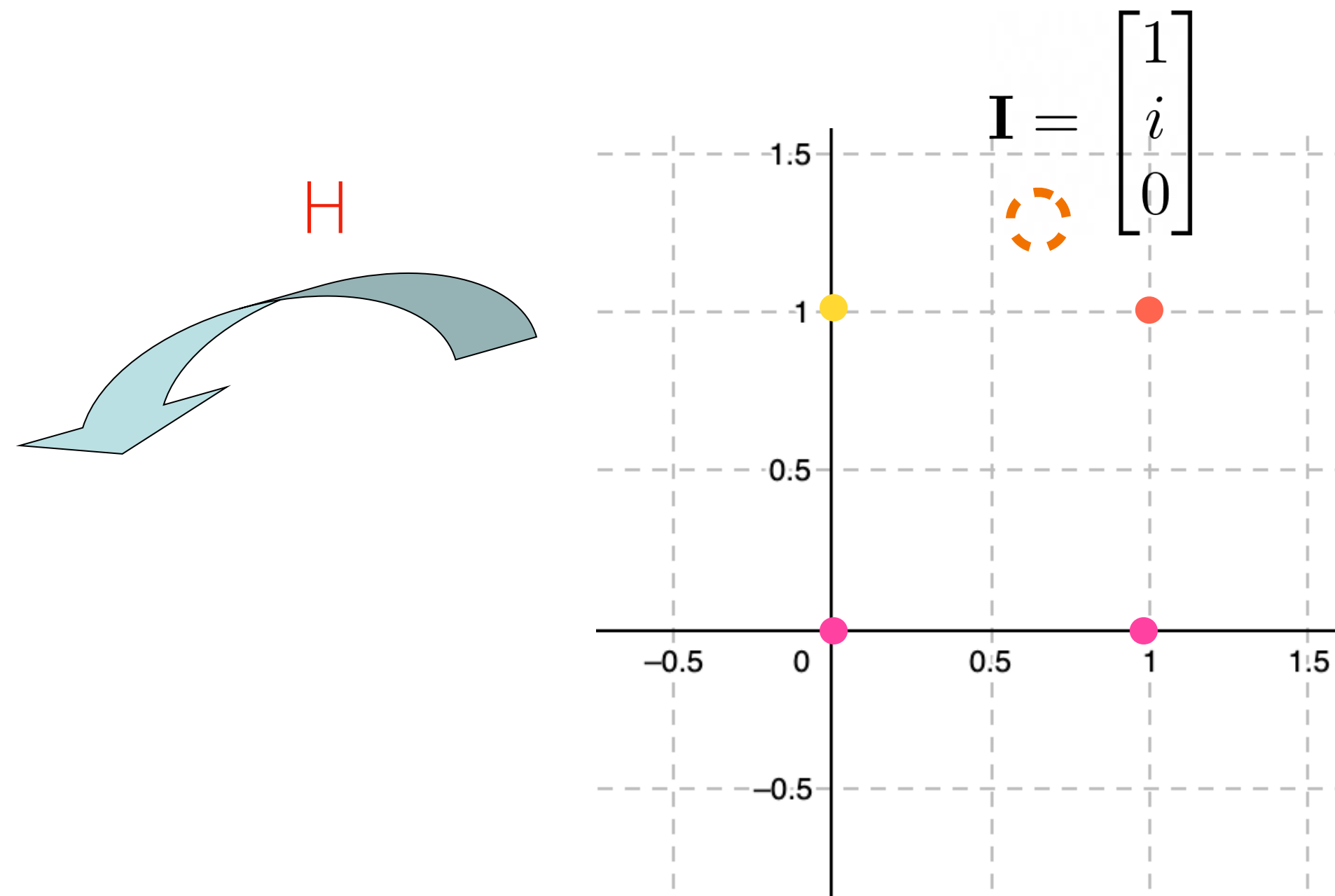
But how to compute?

Image of the Absolute Conic (IAC): ω



Given image of a square

HI (and HJ) lie on the IAC !



$$H = [\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3]$$

$$(\mathbf{h}_1 \pm i\mathbf{h}_2)^T \omega (\mathbf{h}_1 \pm i\mathbf{h}_2) = 0$$

$$\mathbf{h}_1^T \omega \mathbf{h}_2 = 0$$

$$\mathbf{h}_1^T \omega \mathbf{h}_1 = \mathbf{h}_2^T \omega \mathbf{h}_2$$

Identify images of circular points of some plane
The image should lie on IAC

Image of the Absolute Conic (IAC): ω



Given image of a square:
Compute $\mathbf{H}$ that metrically un-
rectifies it

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3]$$

$$\begin{aligned}\mathbf{h}_1^T \omega \mathbf{h}_2 &= 0 \\ \mathbf{h}_1^T \omega \mathbf{h}_1 &= \mathbf{h}_2^T \omega \mathbf{h}_2\end{aligned}$$

If multiple squares are imaged,
each can give a corresponding $\mathbf{H}$

$$\mathbf{H}_a, \mathbf{H}_b, \mathbf{H}_c$$

We get two constraints from each

Q: why not just use the fact that the edges of
the squares are perpendicular lines in 3D?

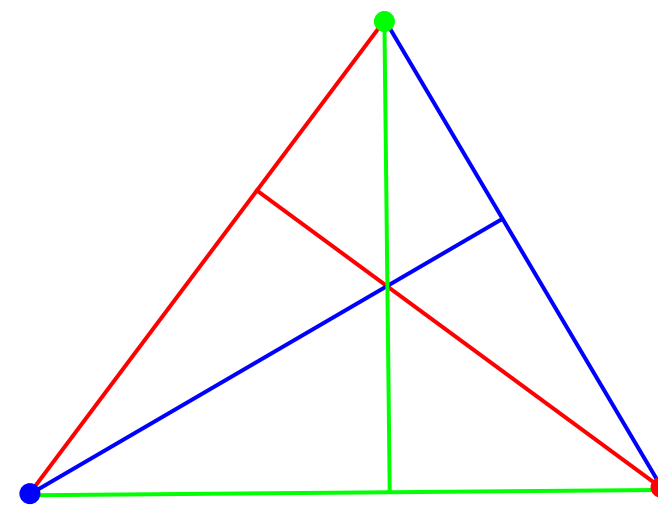
Only yields one constraint

Image of the Absolute Conic (IAC): ω

How can we determine IAC?

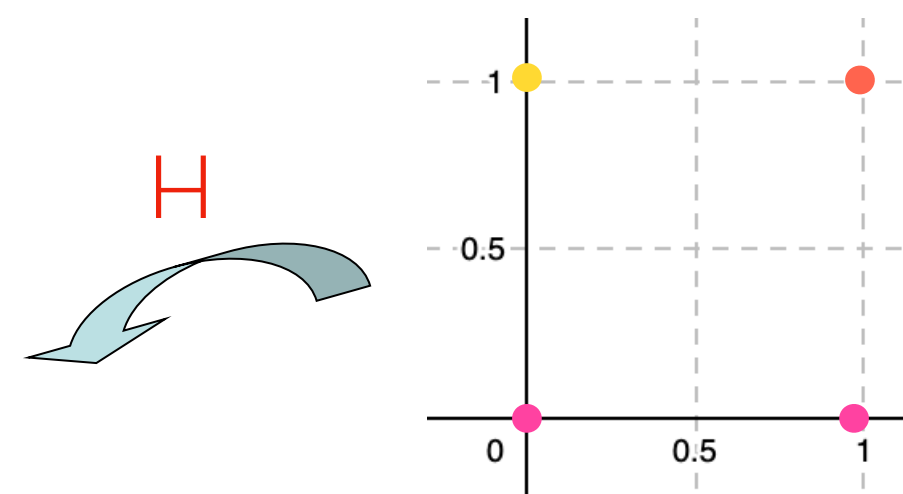
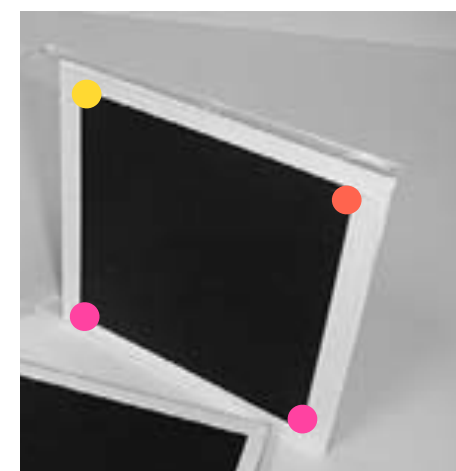
5 constraints needed

Option 1:



$$\mathbf{v}_i^T \omega \mathbf{v}_j = 0$$

Option 2:



$$\mathbf{h}_1^T \omega \mathbf{h}_2 = 0$$

$$\mathbf{h}_1^T \omega \mathbf{h}_1 = \mathbf{h}_2^T \omega \mathbf{h}_2$$

Option 3:

Plane perpendicular to direction (e.g. horizon and vertical direction)

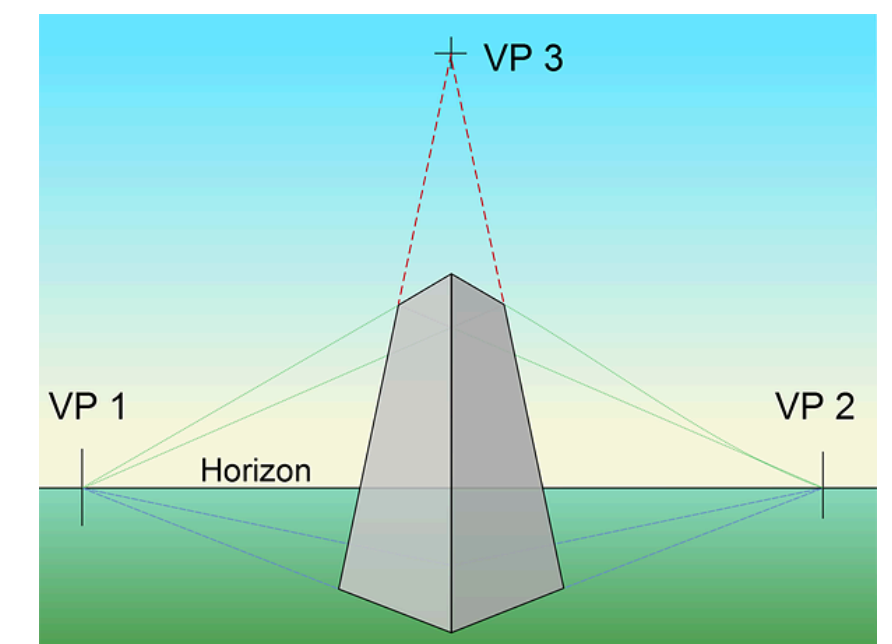
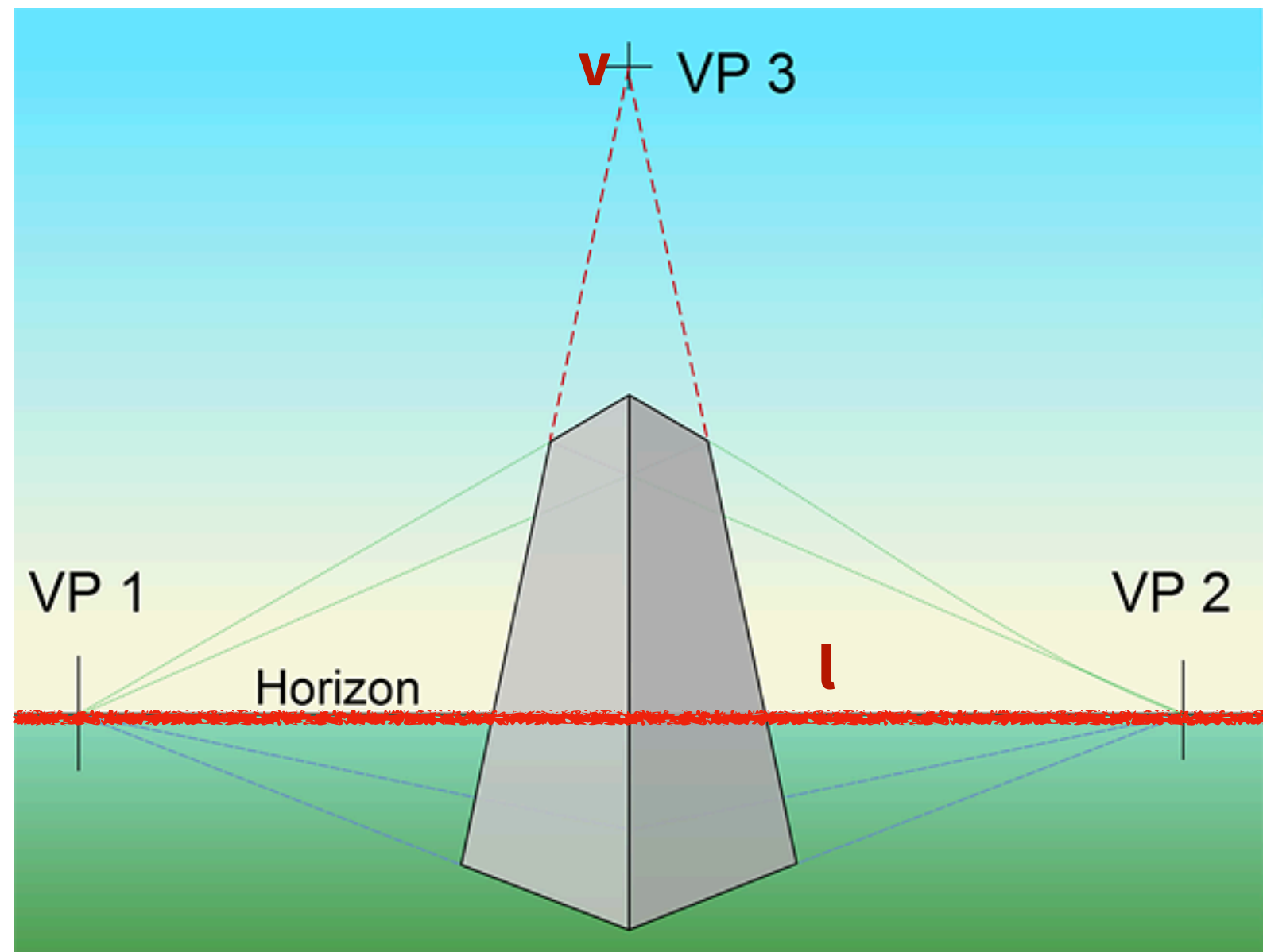


Image of the Absolute Conic (IAC): ω



If $\mathbf{l}$ is the vanishing line of some plane

And $\mathbf{v}$ is vanishing point for an orthogonal direction to the plane

$$\forall \mathbf{v}' \in \mathbf{l}, \mathbf{v}'^T \omega \mathbf{v} = 0 \quad \text{why?}$$

$$\mathbf{l} = \omega \mathbf{v}$$

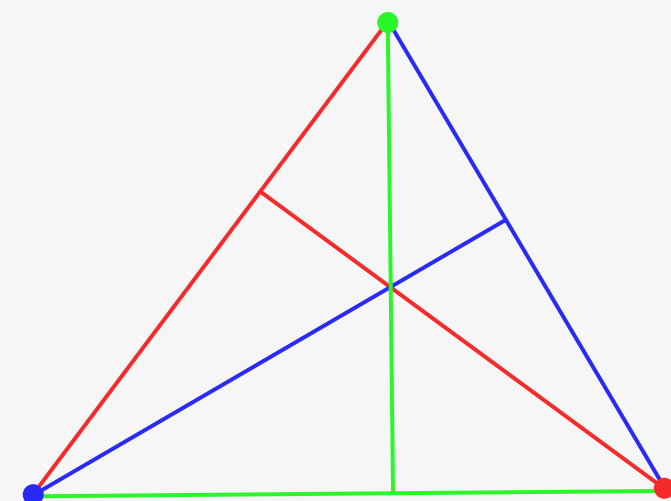
$$[\mathbf{l}]_{\times} \omega \mathbf{v} = \mathbf{0}$$

Q: How many constraints?

Image of the Absolute Conic (IAC): ω

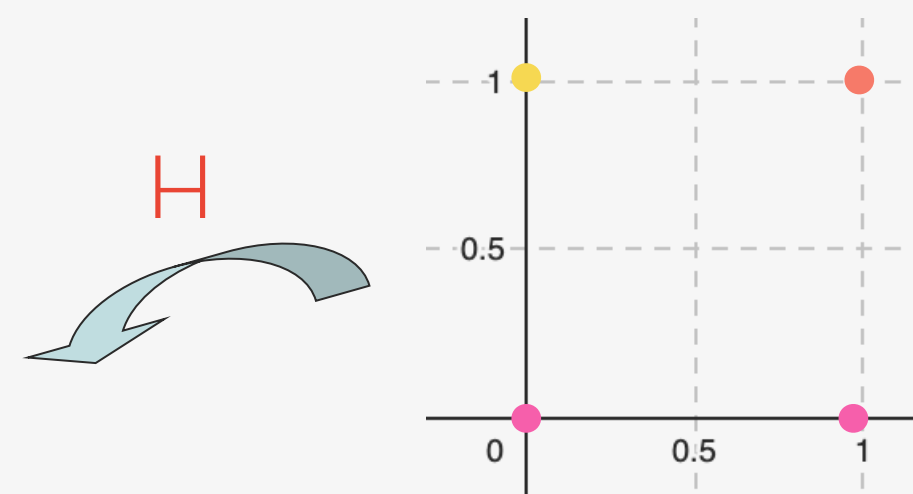
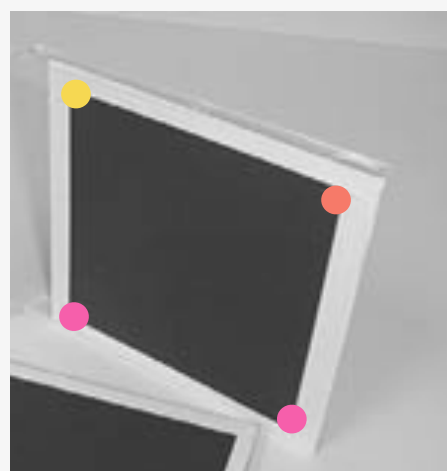
Constraints from Annotation

Option 1:



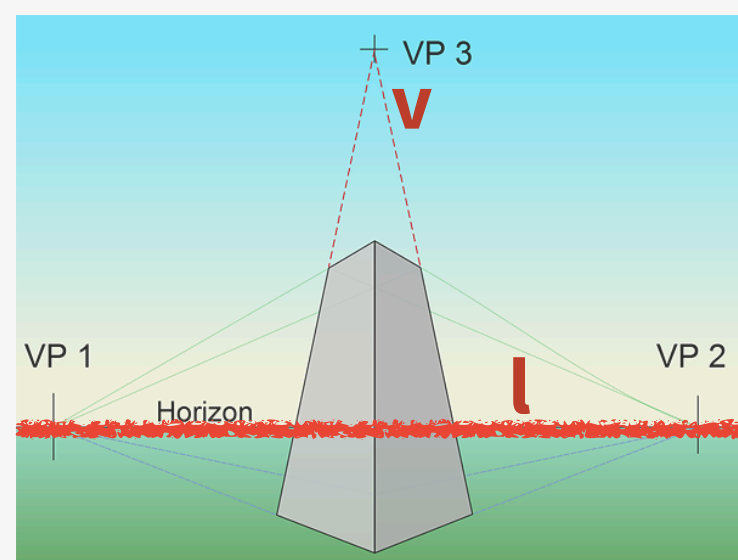
$$\mathbf{v}_i^T \omega \mathbf{v}_j = 0$$

Option 2:



$$\begin{aligned} \mathbf{h}_1^T \omega \mathbf{h}_2 &= 0 \\ \mathbf{h}_1^T \omega \mathbf{h}_1 &= \mathbf{h}_2^T \omega \mathbf{h}_2 \end{aligned}$$

Option 3:



$$[1]_{\times} \omega \mathbf{v} = 0$$

Intrinsic Constraints

1. Zero skew

$$\omega_{12} = \omega_{21} = 0$$

2. Square pixels

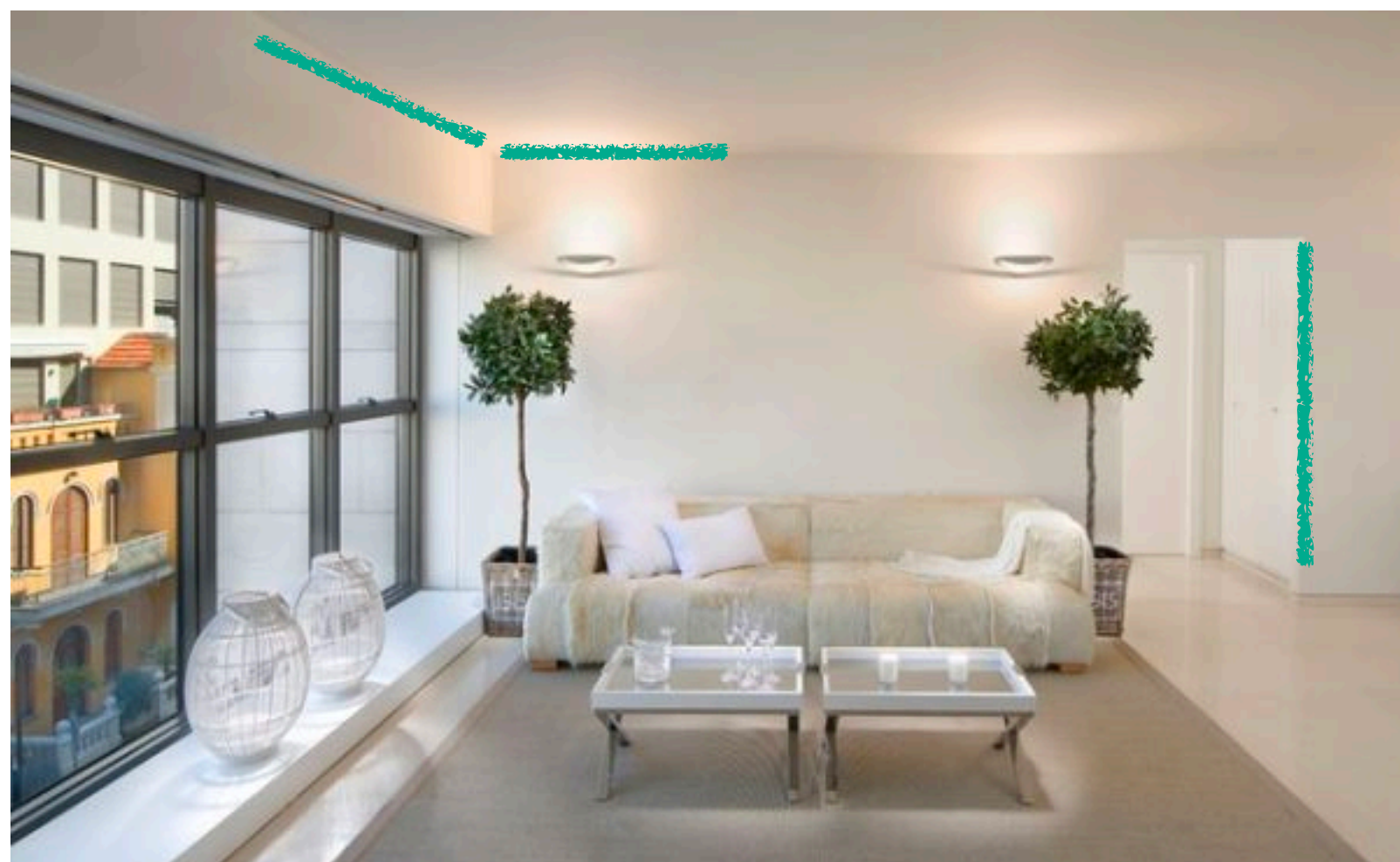
$$\omega_{12} = \omega_{21} = 0$$

$$\omega_{11} = \omega_{22}$$

Calibration — Determining the Intrinsic Matrix

$$\mathbf{P} = \mathbf{K} \mathbf{R} \begin{bmatrix} \mathbf{I} & -\tilde{\mathbf{C}} \end{bmatrix}$$

Possible to recover intrinsics from annotation!



Step 1: Compute ω

Q: if square pixels, how many pairs of perpendicular vanishing points are needed?

Q: In the general case, how many squares should be imaged?

Step 2: Determine $\mathbf{K}$ via Cholesky decomposition

$$\mathbf{A} = \mathbf{L}\mathbf{L}^T \quad \omega = \mathbf{K}^{-T}\mathbf{K}^{-1}$$

Q: Why is ω positive-semi definite?

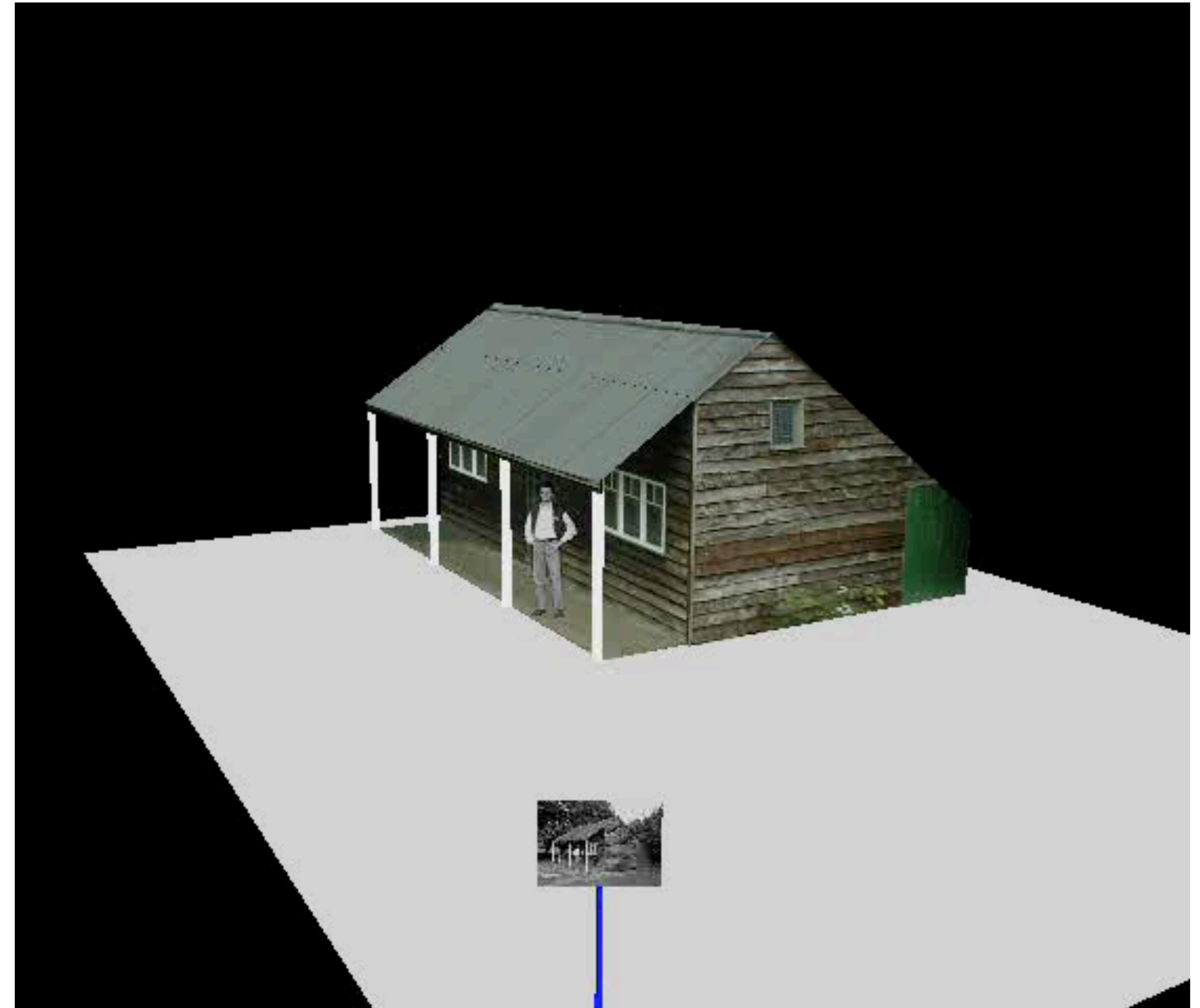
Single-view Reconstruction



Disclaimer: requires lots of
annotations!



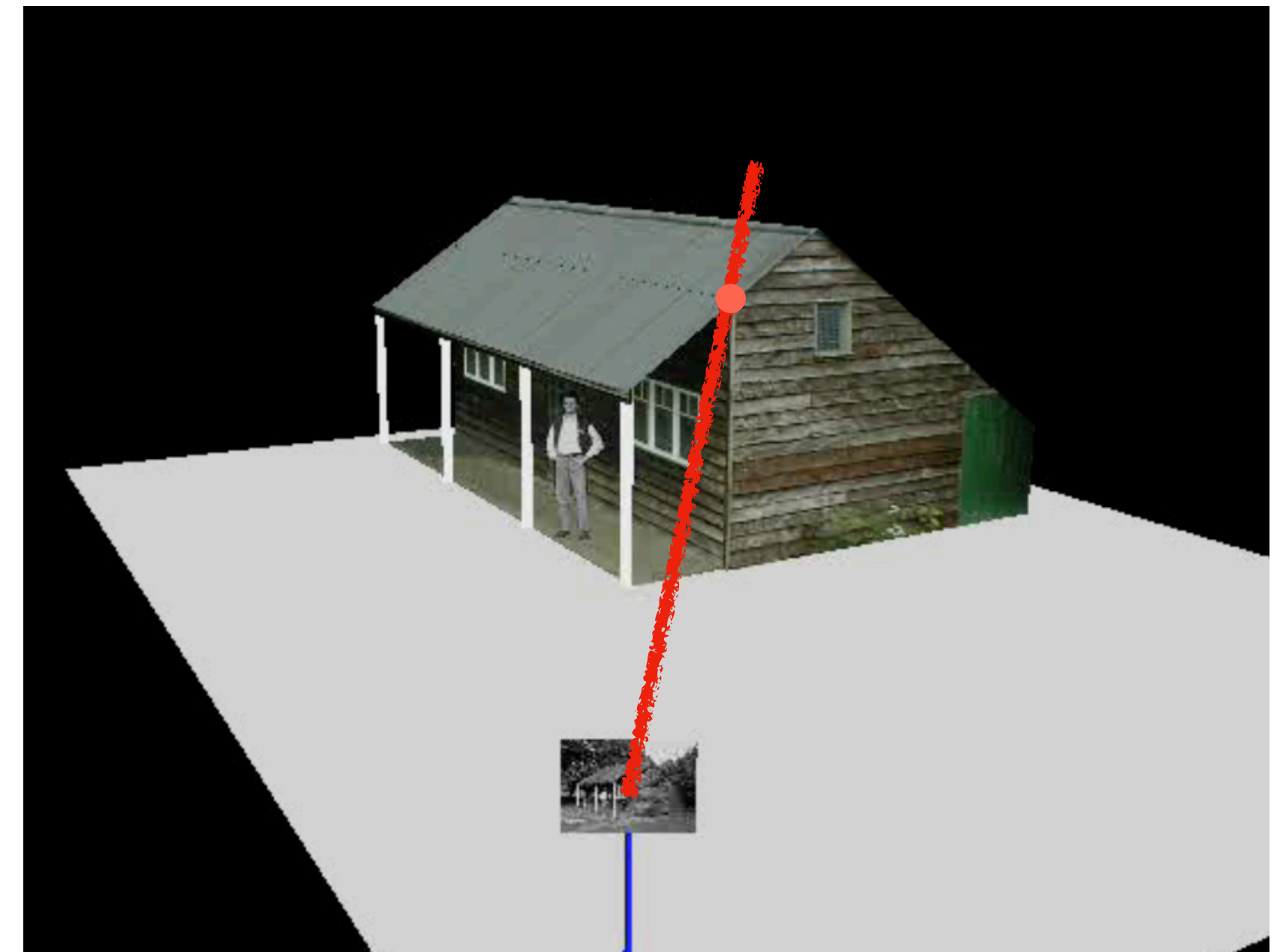
Single-view Reconstruction



Single-view Reconstruction (in Camera frame)



Assumption: K is known



Unique ray corresponding to each pixel

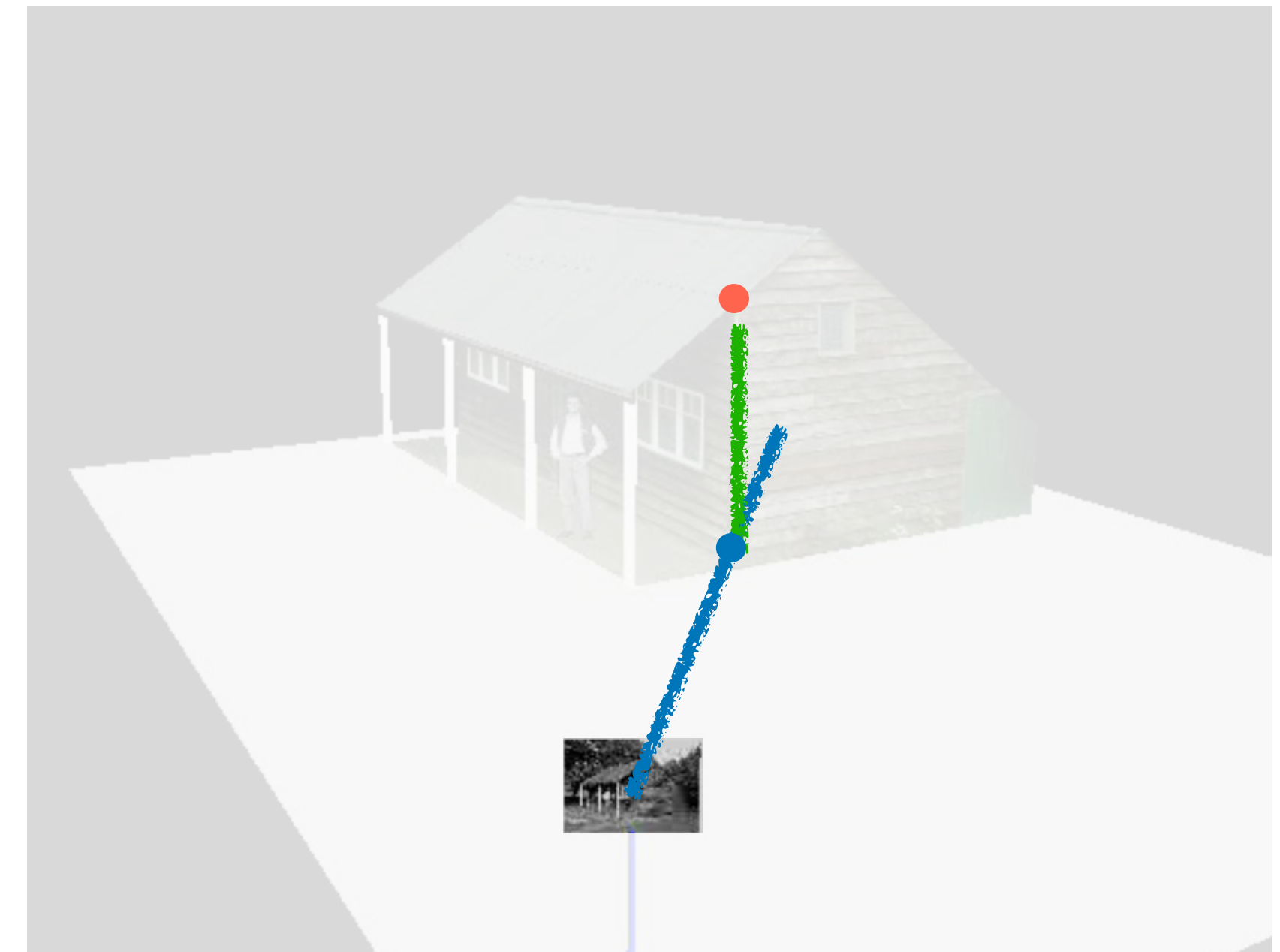
But overall scale of scene is unknown

Assume depth of 1 point

Single-view Reconstruction: Lines



Can we determine orientation of the green line?



Need additional constraint to identify point along ray

Compute vanishing point $\mathbf{v}$ $\rightarrow$ get direction $\mathbf{d} = K^{-1} \mathbf{v}$

Find intersection (or closest crossing) of the blue and green lines in 3D

Single-view Reconstruction: Planes



Lies on same plane as red point

Compute vanishing points for
2 sets parallel lines on a plane



$$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2$$
$$\mathbf{d}_i = K^{-1} \mathbf{v}_i$$



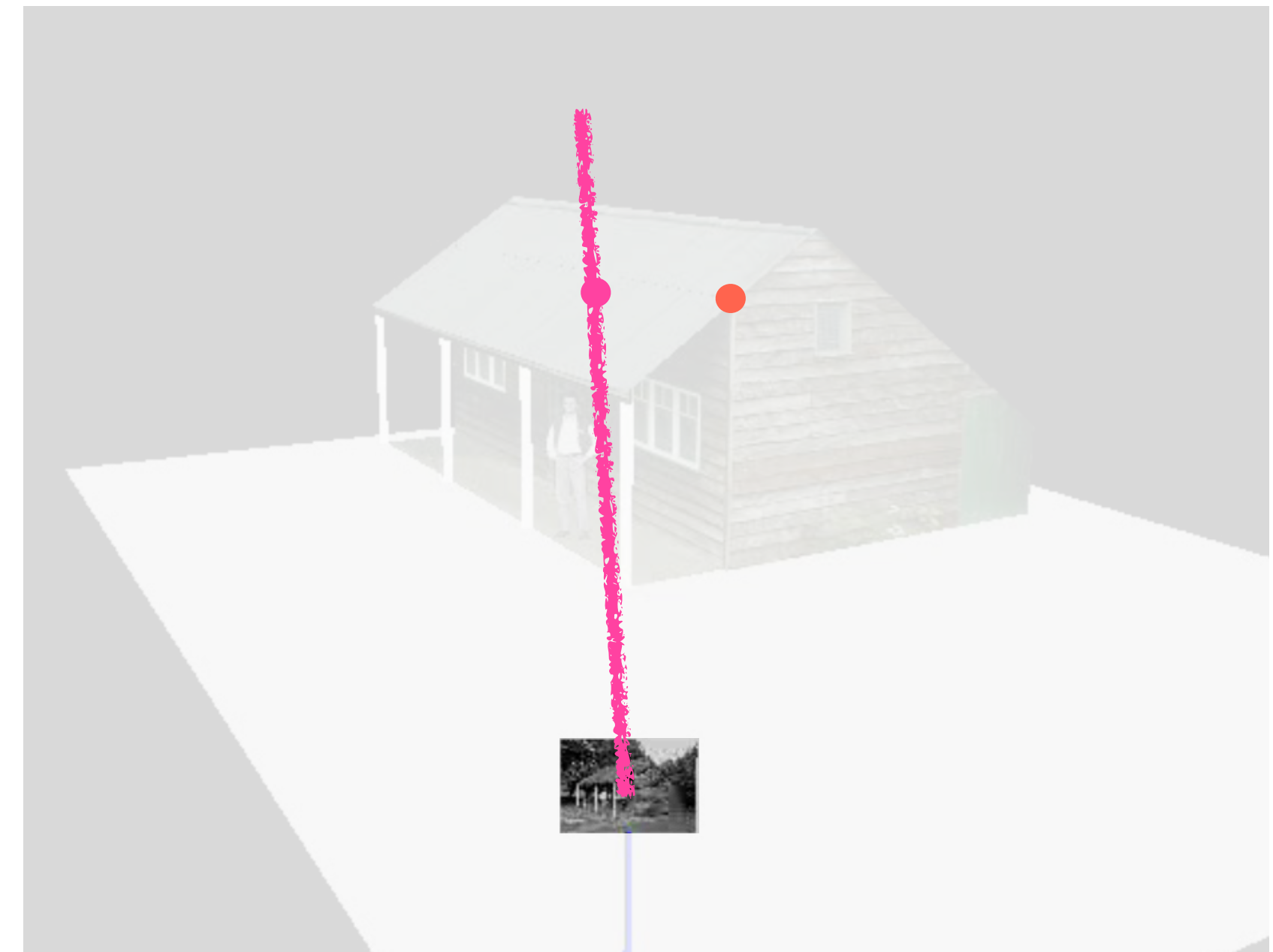
Compute ray-plane
intersection

Plane Equation:

$$\mathbf{n}^\top \tilde{\mathbf{X}} + a = 0$$

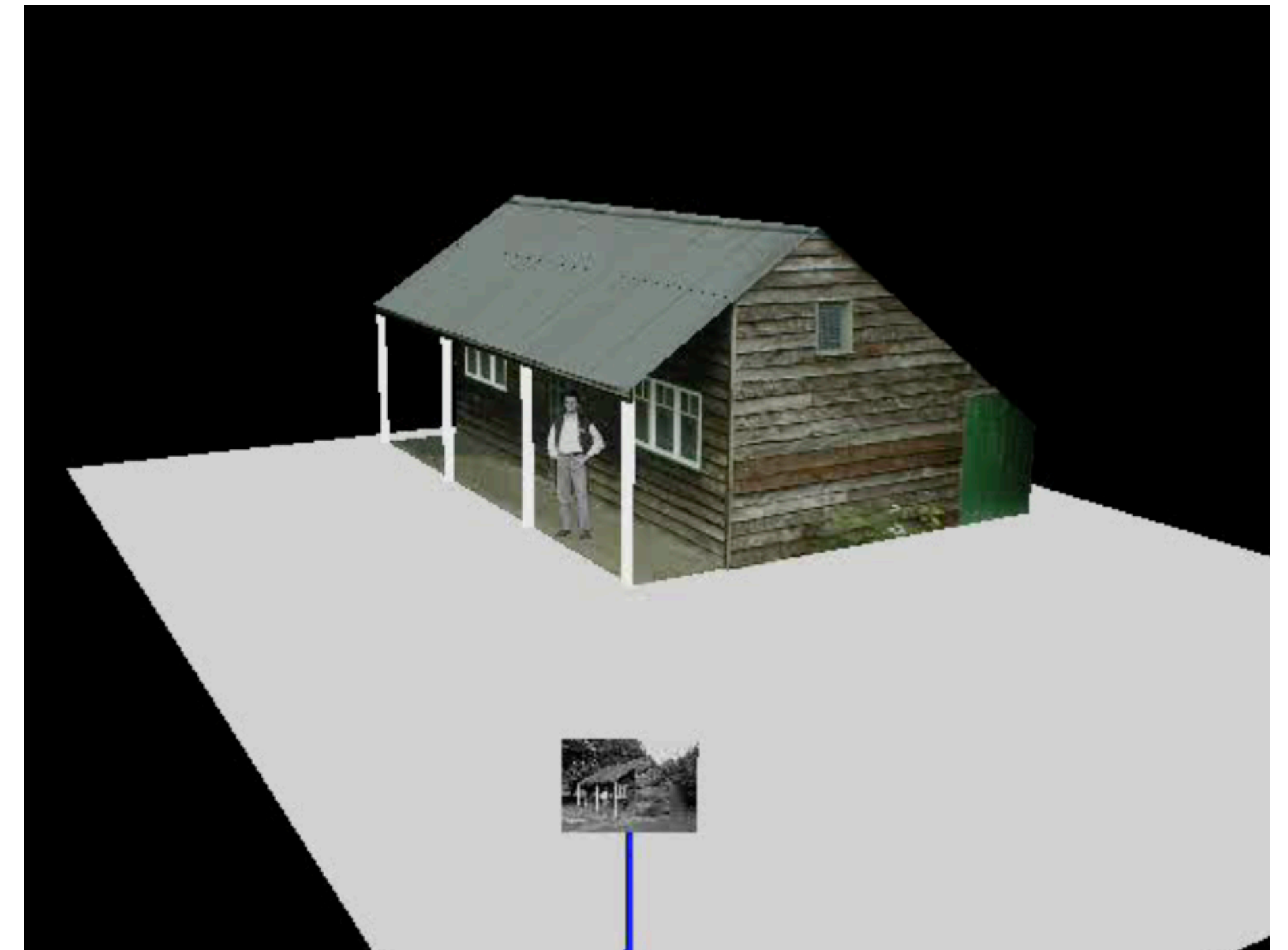
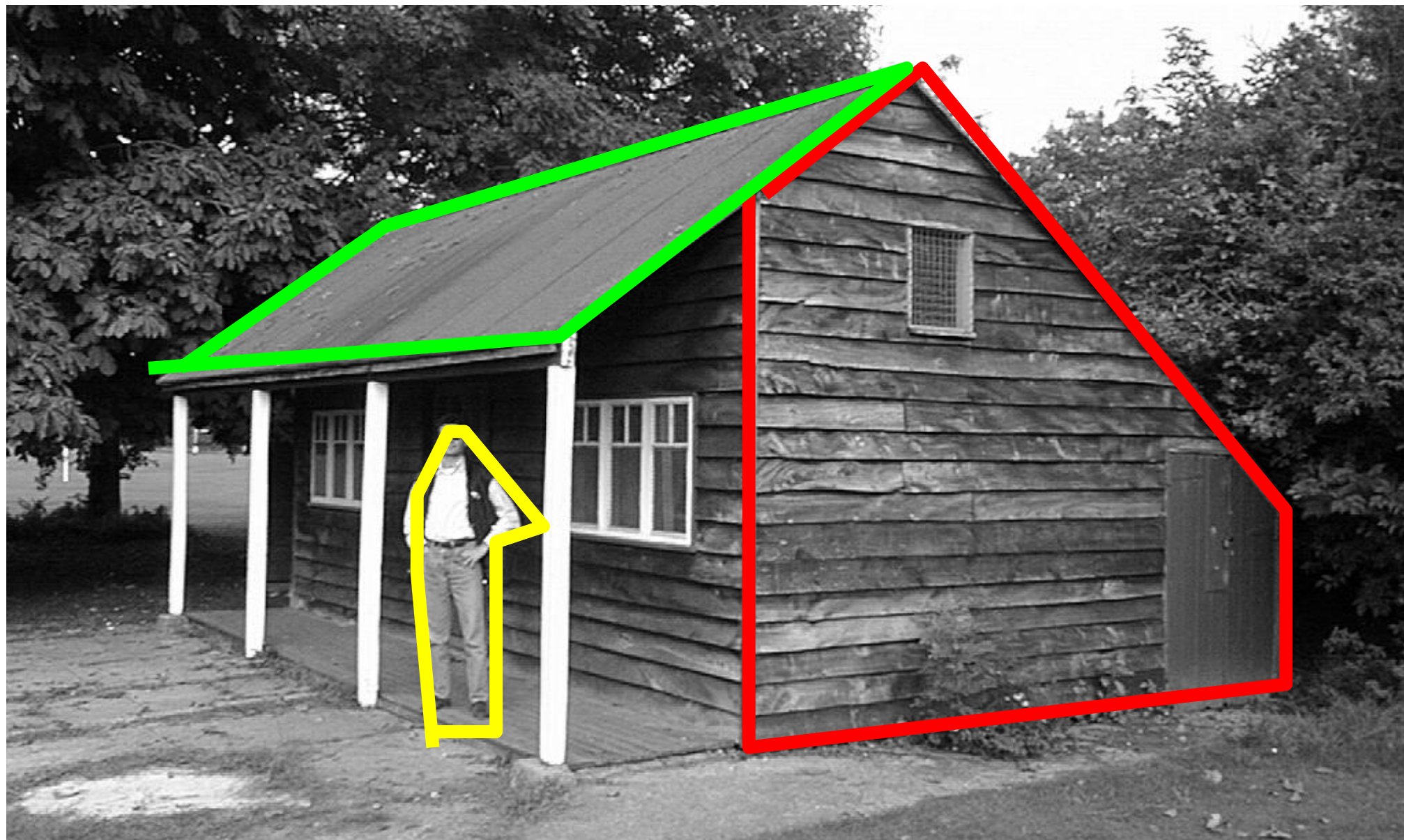
Given $\mathbf{n}$, can compute a
using known red point

**How to get plane
orientation?**



Need additional constraint to identify
point along ray

Single-view Reconstruction: Piecing it Together



Manually annotate planar boundaries,
plane intersections, parallel lines,
occlusions etc..

Lightweight version of this in Assignment 2

Questions?